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Assessing Mental Health from audio using Automatic Speech Recognition

BACHELOR'S THESIS

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STUDENT DECLARATION

I, *Dilnaz Zholmaganbet*, the undersigned, hereby declare that the present BSc thesis work has been prepared by myself and without any unauthorized help or assistance. Only the specified sources (references, tools, etc.) were used. All parts taken from other sources word by word, or after rephrasing but with identical meaning, were unambiguously identified with explicit reference to the sources utilized. In the Appendix, I have clearly indicated whether I used artificial intelligence tools in preparing the thesis; if so, I have specified the manner and extent of their use in the table. I acknowledge that I bear full responsibility for any content generated by artificial intelligence – regardless of the extent of its use.

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Budapest, 20 December 2025

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Dilnaz Zholmaganbet

Abstract

The primary objective of this thesis work was to evaluate the feasibility of a fully automated mental health screening system using clinical speech and contemporary Automatic Speech Recognition (ASR) technology. Given the global shortage of mental health professionals and the limitations of subjective assessment methods, scalable tools capable of objective, early detection are urgently needed. In this context, speech serves as a valuable behavioral biomarker, with linguistic content providing strong indicators of psychological states. A complete audio-to-assessment pipeline was implemented, processing clinical interview recordings using the NVIDIA NeMo ASR system, with the resulting transcripts analyzed for depression risk using a Support Vector Machine classifier, which demonstrated the strongest performance among four evaluated classification models. The central research question was whether ASR transcription errors would significantly degrade screening accuracy. The results show that the automated pipeline achieved 71.4% classification accuracy while maintaining 83.3% sensitivity in identifying "At-Risk" individuals, demonstrating that general-purpose ASR can support scalable preliminary screening with only a minimal performance trade-off. This work concludes that audio-based automated assessment is a viable and promising approach for deployment in telehealth and remote monitoring platforms.

Kivonat

A dolgozat elsődleges célja egy teljesen automatizált mentális egészség-szűrő rendszer megvalósíthatóságának értékelése volt, amely klinikai beszédet és modern automatikus beszédfelismerő (ASR) technológiát alkalmaz. A világméretű mentálhigiénés szakemberhiány és a szubjektív értékelési módszerek korlátai miatt sürgősen szükség van skálázható, objektív és korai felismerést lehetővé tevő eszközökre. Ebben a kontextusban a beszéd értékes viselkedési biomarkerként szolgál, melynek nyelvi tartalma erős jelzőt nyújt a pszichológiai állapotokra. Egy teljes hang-alapú értékelési folyamatot valósítottunk meg, amely a klinikai interjúfelvételeket az NVIDIA NeMo ASR rendszerrel dolgozza fel, majd az így kapott átiratokat egy Támogató Vektorgép (SVM) osztályozóval elemzi depresszió kockázat szempontjából – ez a modell mutatta a legjobb teljesítményt a négy tesztelt osztályozó közül. A központi kutatási kérdés az volt, hogy az ASR-átirati hibák jelentősen rontják-e a szűrés pontosságát. Az eredmények szerint az automatizált folyamat 71,4%-os osztályozási pontosságot ért el, miközben 83,3%-os érzékenységet (szenzitivitást) tartott fenn a „rizikós” egyének azonosításában, ami azt igazolja, hogy az általános célú ASR képes támogatni a skálázható előszűrést csupán minimális teljesítménycsökkenés árán. Ez a tanulmány arra a következtetésre jut, hogy a hangalapú automatizált értékelés egy életképes és ígéretes megközelítés a telemedicina és a távoli monitorozási platformok számára.

Chapter 1

1 Introduction

Mental health conditions, including depression and anxiety disorders, are among the most common health concerns worldwide, affecting hundreds of millions of people and contributing significantly to the global burden of disease [1]. These conditions can profoundly impact an individual's ability to function in daily life, affecting work, education, and personal relationships [2]. The associated economic costs are substantial, impacting both healthcare systems and broader economic productivity [3]. Additionally, mental health disorders are closely linked to elevated suicide risk, representing a critical public health priority [4].

Early detection and intervention are crucial for improving outcomes and reducing long-term disability [5]. However, current diagnostic practices often rely on subjective methods such as self-reported questionnaires and clinical interviews, which can be influenced by recall bias, fluctuating mood, and social stigma [6]. These challenges are compounded by a severe global shortage of mental health professionals, creating significant barriers to timely and accessible assessment, especially in underserved regions [7].

Technology-assisted assessment offers a promising avenue for scalable and objective mental health screening. A growing body of research indicates that vocal characteristics can serve as behavioral markers of psychological states. These characteristics include lexical content, tone, speech rate, and pausing patterns. [6]. For instance, specific conditions like depression are associated with identifiable speech patterns, such as increased use of negative and self-referential language [8, 9] and reduced variation in vocal pitch [10, 11]. This evidence suggests that audio recordings from clinical or natural conversations could form a valuable data source for automated screening tools.

Automatic Speech Recognition (ASR) technology, which converts spoken language into text, has advanced to a level of accuracy suitable for many practical applications [12]. In healthcare, ASR enhanced by natural language processing is being explored to automate clinical documentation, allowing clinicians to dictate notes and summarize patient-provider conversations to populate health records and support clinical decisions [12]. If such technology can reliably transcribe the nuanced speech found in clinical settings, it could enable automated, low-cost mental health screening through telehealth platforms, mobile applications, and remote monitoring systems [13, 14].

However, a key practical challenge lies in the complexity of clinical speech. Therapeutic dialogues are characterized by emotional expression, disfluencies, pauses, and specialized vocabulary, all of which can increase transcription error rates [10]. Inaccurate transcripts may compromise downstream analysis, potentially leading to missed cases or false-positive screenings [11].

Therefore, the central aim of this thesis work is to evaluate whether an ASR technology can transcribe clinical speech accurately enough to support a reliable, general mental health screening tool. A complete audio-to-assessment pipeline was developed, which processes clinical interview recordings using the NVIDIA NeMo ASR system and analyzes the resulting text for mental health risk indicators, with depression serving as a primary case study. The system's performance is evaluated by comparing screening outcomes based on automated transcripts against those derived from manual transcripts. This work seeks to quantify the impact of transcription errors on screening accuracy and to assess the practical feasibility of deploying audio-based mental health assessment at scale.

1.1 Thesis Contributions

The primary contribution of this thesis is a comprehensive evaluation of the feasibility of fully automated, audio-based mental health screening using contemporary Automatic Speech Recognition (ASR) technology. Specific contributions include:

- An empirical assessment of ASR performance on clinical speech, quantifying its accuracy and identifying characteristic error patterns in therapeutic dialogue.
- The establishment of a performance baseline for text-based mental health classification using clinical transcripts, identifying an optimal approach.
- A direct quantification of how ASR transcription errors impact classification accuracy, isolating the effect of automation on screening performance.
- A clinically informed error analysis that reveals the conditions under which the automated pipeline succeeds or fails, providing practical insights for deployment.
- Evidence-based conclusions on the current viability of ASR for scalable mental health screening, outlining the trade-off between automation and diagnostic validity.

1.2 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the literature on speech-based mental health assessment, covering acoustic and linguistic approaches, automatic speech recognition applications in healthcare, and text-based mental health detection methods. Chapter 3 describes the methodology in detail, including the DAIC-WOZ dataset, preprocessing procedures, the NVIDIA NeMo ASR system, TF-IDF feature extraction, classification algorithms, and experimental design. Chapter 4 presents the experimental results, evaluating ASR performance on clinical audio, establishing baseline classification performance with manual transcripts, testing the complete audio-to-assessment pipeline, analyzing linguistic features, and discussing key findings. Chapter 5 concludes with a summary of findings, discusses implications for audio-based mental health screening, and suggests directions for future research including multimodal approaches integrating acoustic-prosodic with linguistic analysis.

Chapter 2

2 Related Work

Speech has emerged as a valuable, objective source of biomarkers for mental health, functioning as a dual-channel signal where phonetics captures physiological states and semantics reveal psychological conditions [15]. This chapter reviews prior research on mental health assessment from speech, examining both acoustic and linguistic approaches, and identifies the specific research gap this thesis addresses.

Mental health conditions produce observable changes in speech patterns, making audio analysis a long-standing avenue for computational assessment. Historically, research established strong correlations between vocal acoustics and clinical states like depression [16]. Acoustic features capture how speech is produced, including prosodic characteristics such as pitch variability, speech rate, and pause duration, as well as voice quality measures like fundamental frequency (F0), jitter, shimmer, and harmonics-to-noise ratio. The physiological basis for these changes is well-established: speech production engages more motor fibers than any other human mechanical activity [17], making it particularly sensitive to physiological and cognitive alterations. Even slight changes in mental state can produce noticeable acoustic variations [18]. Depression, for instance, has been associated with monotone speech, slower articulation, and prolonged silences [10]. Clinicians have long characterized depressed speech as dull, lifeless, and lacking emotional variation [19], and studies conducted in the 1970s demonstrated that listeners could perceive noticeable differences in prosodic characteristics in depressed patients before and after treatment [20].

Researchers have achieved promising classification results using carefully selected acoustic-phonetic parameters. One study demonstrated 86% accuracy in distinguishing healthy from depressed speech using Support Vector Machine (SVM) classifiers on Hungarian speech data, emphasizing the importance of feature selection and optimization [21]. Earlier research examined segmental parameters of depressed speech, finding significantly higher jitter and shimmer values and lower first and second formant frequencies in depressed individuals compared to normal speech [22]. Studies have found significant correlations between depression severity and pause-related measures, indicating that depressed individuals have trouble in word selection and speech planning [23, 24]. More recently, research has demonstrated that pre-trained deep learning models like Wav2vec 2.0 can automatically learn optimal features from raw audio data and achieve high efficacy in depression recognition [25].

However, despite these successes, acoustic approaches face fundamental limitations. The reliability of even advanced acoustic features is questionable. Research has demonstrated that prosodic features like F0 are too coarse to capture subtle vocal fold tension, finding that glottal parameters provided more discriminatory information [26]. Acoustic features are inherently variable, susceptible to background noise, and often represent correlational rather than causal links to pathology, leading to challenges with specificity and robustness [10]. Furthermore, acoustic-based approaches face critical practical challenges related to data requirements. Studies have noted that robust acoustic analysis requires substantially larger databases of depressed speech samples than are typically available, limiting the generalizability of findings from small-scale studies [22]. This data scarcity issue is particularly problematic because effective acoustic models demand extensive clinical audio datasets with diverse speakers, recording conditions, and symptom presentations. But it remains difficult to obtain these kinds of resources due to privacy concerns and the logistical challenges of clinical data collection. Most critically, model ablation studies in multimodal research consistently demonstrate that text-based linguistic features contribute more to diagnostic accuracy than acoustic features alone, with acoustics sometimes providing only marginal gains or even worsening performance compared to linguistic

models [27, 28].

The limitations of acoustic approaches have motivated a decisive shift toward text-based analysis, which recognizes that audio contains not only acoustic information but also linguistic content that can be extracted and analyzed. Text-based analysis examines the semantic and syntactic content of speech through transcripts, which provide a stable, reviewable artifact that directly captures what is said rather than just how it is said. Research has demonstrated that text transcription provides an explicit and persistent record of clinical dialogue, making it one of the best candidates for distress analysis tasks [9, 29]. Unlike ephemeral acoustic signals, transcripts allow for the application of sophisticated Natural Language Processing (NLP) techniques to identify nuanced psychological cues embedded within linguistic structure, such as patterns of emotional expression.

The feature extraction stage is equally critical when working with linguistic features. In text-based classification tasks, words are converted into numeric features using methods such as TF-IDF (Term Frequency-Inverse Document Frequency), which assigns weights to words based on their frequency in a specific document and their uniqueness across the entire document corpus [30]. Researchers have achieved effective mental health classification using both traditional feature-engineering methods like TF-IDF and modern deep learning models. Studies have demonstrated the continued relevance of TF-IDF, achieving 95.33% accuracy in a binary mental health classification task on social media text using a TF-IDF vectorizer with a Support Vector Classifier [31]. Similarly, research has successfully applied TF-IDF feature extraction for mental health detection [32]. These "bag-of-words" methods, while computationally efficient and interpretable, provide strong performance baselines by capturing key term prevalence, firmly establishing text as a robust medium for mental health assessment.

Automatic Speech Recognition (ASR) technology provides the crucial bridge that allows powerful text-based linguistic analysis to be applied directly to audio recordings. Modern, pre-trained ASR models, particularly deep learning-based architectures like wav2vec2.0, Whisper, and RNN-Transducers, have achieved remarkable accuracy on diverse speech data. Recent research has demonstrated that pre-trained ASR models can transcribe psychiatric interviews with a word error rate as low as 7.8% on clinical interview data, making high-fidelity linguistic analysis a practical reality [33]. This level of accuracy has motivated increasing interest in applying ASR to clinical contexts, where it promises to enable scalable, automated mental health screening by converting audio from phone calls, video consultations, and telehealth sessions into text suitable for NLP analysis.

However, ASR performance in clinical and healthcare settings presents unique challenges that are not reflected in general benchmarks. A comparative evaluation of ASR systems for recognizing Dutch clinical terminology revealed important trade-offs between lexical accuracy and adaptability [34]. Notable error types included phonetic substitutions, semantic approximations, and truncations, each with distinct clinical implications. The study further showed that even with domain-specific training data (12 hours of medical speech), ASR systems continue to struggle with accurately capturing specialized healthcare terminology. These findings highlight the need for task-specific optimization and indicate that further improvements may require more extensive and higher-quality clinical training data to support reliable medical ASR applications [34]**.** Recent work demonstrated both the potential and limitations of applying ASR in clinical populations [35]. The study fine-tuned a wav2vec2.0 XLS-R model to recognize pronunciation patterns in children with speech sound disorders, achieving high agreement with human annotations. In contrast, the pre-trained Whisper model, although showing strong performance on general Korean speech recognition tasks, exhibited clear limitations when applied to speech from children with SSDs. This comparison highlights that general-purpose ASR models trained on standard speech may not perform reliably on atypical clinical speech without domain-specific adaptation.

Mental health populations present related but distinct challenges for automatic speech recognition. One study evaluated ASR performance on speech samples from 50 individuals with schizophrenia-

spectrum disorders and reported word error rates ranging from 0.31 to 0.58 [36]. Importantly, ASR performance varied systematically based on country of birth and the severity of positive symptoms, indicating that recognition accuracy is not uniform across clinical populations. Further analysis showed that ASR-generated transcripts exhibited significantly higher semantic similarity scores and fewer sentences than manual transcripts, as well as weaker correlations between NLP-derived metrics and clinical symptom measures. These findings highlight that evaluating ASR in clinical settings requires looking beyond word error rate alone, as the impact of recognition errors depends not only on their frequency but also on their type, meaning, and context. Although individuals with mental health conditions may not exhibit speech sound disorders or use specialized medical terminology, their speech can differ from typical ASR training data in ways that affect downstream NLP analyses.

This confluence of factors defines the core research gap that this thesis addresses. Automatic speech recognition has shown value for transcribing general medical interviews, and substantial research has examined mental health detection from audio using both audio features and manual transcripts. However, the practical feasibility of a complete automated system, which converts raw clinical audio to text using ASR and then performs mental health classification, has not been sufficiently explored. Therefore, the central question is whether ASR-introduced transcription errors significantly degrade the performance of text-based mental health classifiers when applied to clinical interviews with individuals experiencing mental health challenges. Although larger, high-quality clinical audio datasets with manual transcripts would enable fine-tuning of ASR models specifically for mental health applications, such datasets remain scarce. Consequently, it remains unknown whether current general-purpose ASR models offer sufficient accuracy for mental health screening, or whether the efficiency gains of automation are offset by a loss in diagnostic validity, creating a significant barrier.

Chapter 3

3 Methodology

This chapter provides a comprehensive explanation of the multi-stage computational pipeline for mental health assessment introduced in the preceding chapter. The primary goals of this methodology are to show a practical application of Automatic Speech Recognition (ASR) for converting clinical speech into text and to identify the most effective natural language processing models for analyzing the transcribed text. To achieve this, the chapter is structured as follows. It begins with a description of the dataset and preprocessing steps (Section 3.1), followed by a high-level overview of the entire system (Section 3.2). Next two sections delve into the main components of the process, including the ASR module developed using the NVIDIA NeMo deep learning toolkit (Section 3.3), and principles of feature extraction and detailed discussion of the chosen method TF-IDF vectorizer (Section 3.4). Section 3.5 discusses the classification of mental health, while Section 3.6 outlines the training and testing procedures. Finally, Section 3.7 describes the evaluation metrics employed in this study.

3.1 Data Description and Preprocessing

This section outlines the dataset utilized in this study along with the preprocessing steps undertaken to prepare the data for analysis. The quality and preparation of data are vital to the success of any pipeline, especially when dealing with clinical data, which often presents distinct challenges.

3.1.1 DAIC-WOZ Dataset

This research utilized the Distress Analysis Interview Corpus - Wizard of Oz (DAIC-WOZ) dataset to assess the effectiveness of an Automatic Speech Recognition (ASR) system in processing clinical speech and to review the entire mental health evaluation process [37]. As a component of the broader Distress Analysis Interview Corpus, the DAIC-WOZ includes clinical interviews aimed at aiding the diagnosis of mental health conditions such as depression, anxiety, and post-traumatic stress disorder.

The dataset consists of audio recordings, manually created transcripts, and psychological assessment scores. Each participant's depression level is evaluated using the eight-item Patient Health Questionnaire (PHQ-8) [38]. The transcripts include speaker labels, timestamps and the text, as shown in Table *3.1*. A separate CSV file contains each participant's PHQ-8 depression scores for clinical analysis.

start_time	stop_time	speaker	value
35.18	37.99	Ellie	hi i'm ellie thanks for coming in today
38.59	41.72	Ellie	i was created to talk to people in a safe and secure environment
42.46	46.77	Ellie	think of me as a friend i don't judge i can't i'm a computer
47.67	57.06	Ellie	i'm here to learn about people and would love to learn about you i'll ask a few questions to get us started and please feel free to tell me anything your answers are totally confidential
58.43	59.4	Ellie	how are you doing today
59.93	61.36	Participant	i'm fine how about yourself

Table 3.1.Sample of the transcript file: Participant number 302

This well-established clinical dataset offers reliable diagnostic metrics for assessing mental health conditions. In this study, binary classification method was implemented. Participants with scores below 10 were labeled as "Not At-Risk", while those with scores from 10 to 24 (the maximum) were labeled as "At-Risk", indicating potential mental health issues that may require professional intervention [38].

The dataset was strategically utilized to achieve two complementary goals. For the evaluation of Automatic Speech Recognition (ASR) only the test set's audio files and their corresponding manual transcripts were used to evaluate the performance of the pre-trained chosen ASR model on previously unseen clinical data. For mental health classification, the entire set of transcripts was utilized to develop and assess a feature extraction combined with a machine learning models, ensuring robust model training while maintaining appropriate data separation.

Out of the original 192 files, 10 were excluded due to excessive audio static and other technical issues that could compromise both ASR performance and subsequent analysis. The remaining 182 files were split into training (72%), testing (15%), and validation (13%) sets, ensuring there was enough data for a comprehensive evaluation while maintaining clinical realism.

3.1.2 Data Preprocessing Steps

The raw data required comprehensive preprocessing to address several challenges before being suitable for both ASR evaluation and mental health classification. This subsection describes the key preprocessing steps applied to ensure reliable analysis.

For the ASR performance evaluation, only the test set audio files were processed. The audio quality was enhanced through volume normalization to solve inconsistent recording levels while preserving speech quality. A main preprocessing decision involved selecting an appropriate time frame for analysis. As shown in Figure 3.1, analysis of participant speech patterns revealed that conversations

typically begin around 60 seconds, with participants speaking an average of 1,548 words over approximately 15 minutes. However, to ensure consistent evaluation conditions and capture meaningful dialogue while avoiding initial recording artifacts, a window from 120 seconds till the end of recording was selected. This timeframe captures the core conversational content while maintaining practical processing constraints. Also, it was selected based on analysis showing it captures 95% of meaningful participant dialogue while excluding initial recording artifacts.

The same temporal alignment was applied to the corresponding manual transcripts, ensuring direct comparability between ASR-generated and ground truth text. Only participant speech segments were retained, excluding the virtual interviewer's dialogue to focus analysis on clinically relevant participant language patterns.

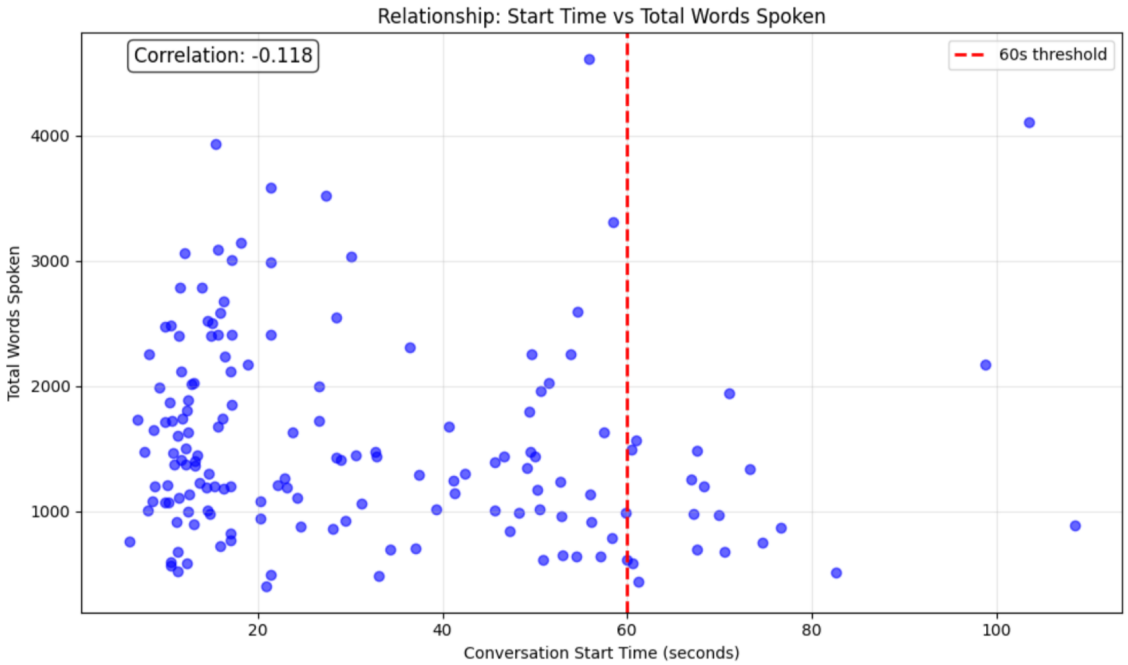


Figure 3.1. Start Time vs Total Words Spoken Relationship of the dataset

For the mental health classification task, transcripts from all 182 files underwent systematic text preprocessing. The processing pipeline included speaker filtering to isolate participant speech segments and temporal alignment using the same time frame established for ASR evaluation. The text was then cleaned to remove non-verbal cues while preserving emotional content, tokenized into individual words. Then the Porter Stemming algorithm was applied to normalize vocabulary by reducing words to their root forms, thereby increasing the precision of semantic analysis and improving the consistency of feature extraction across the clinical transcripts [39]. This approach ensured consistent feature extraction while maintaining clinically relevant linguistic patterns.

As illustrated in Figure 3.2, the dataset exhibits significant class imbalance with approximately twice as many healthy participants (PHQ-8 < 10) compared to those showing depression symptoms (PHQ-8 ≥ 10). This 2:1 ratio reflects real-world clinical prevalence but poses challenges for machine learning model training by making it biased towards the common class. To solve this, complementary class imbalance handling strategy was applied to the minority class during training (more details in Section 3.6), ensuring balanced learning of both mental health conditions without introducing bias toward the majority class.

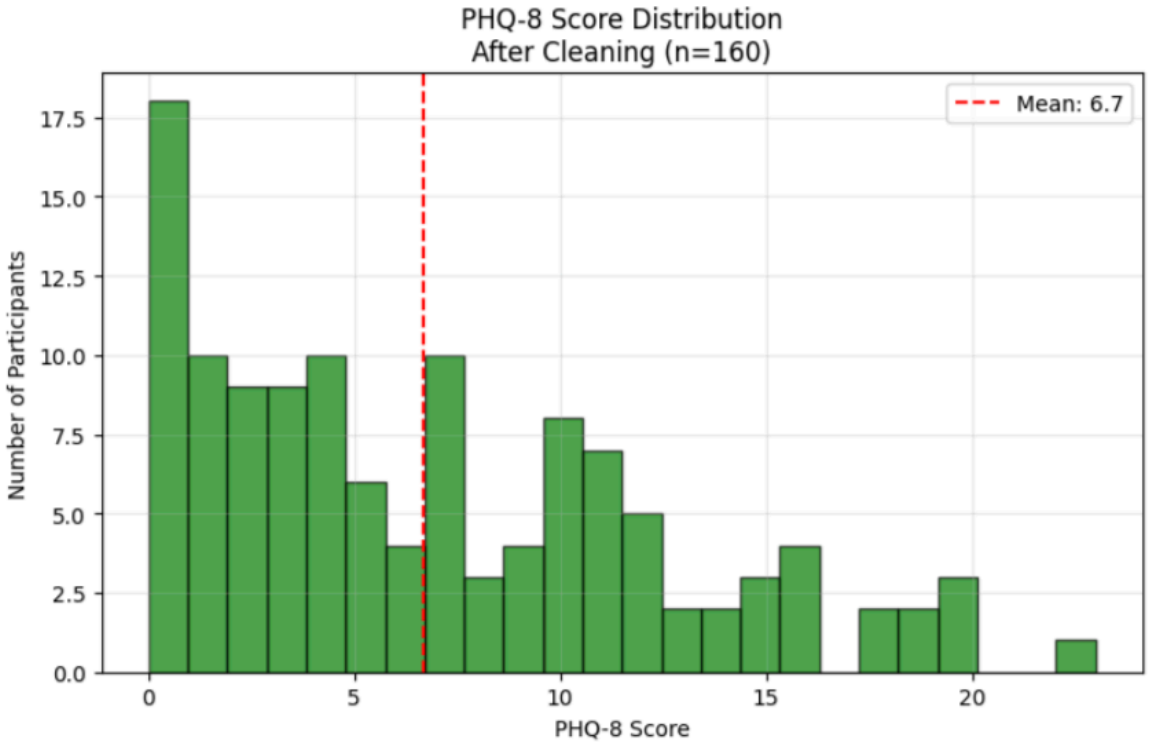


Figure 3.2.PHQ-8 Score Distribution in the Dataset

The data preprocessing pipeline illustrated in Figure 3.3, outlines the sequential steps applied to transform raw clinical interviews into structured data suitable for ASR evaluation and mental health classification.

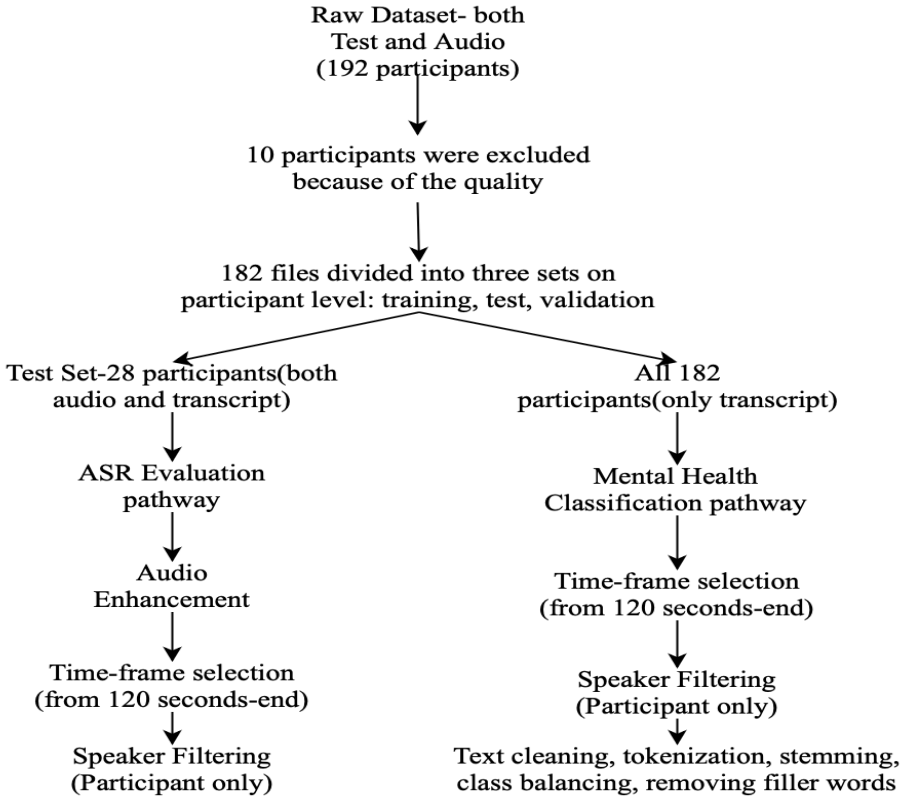


Figure 3.3.Preprocessing Pipeline

3.2 System Architecture Overview

This research implements a comprehensive pipeline for mental health assessment through automated speech processing, as illustrated in Figure 3.4.

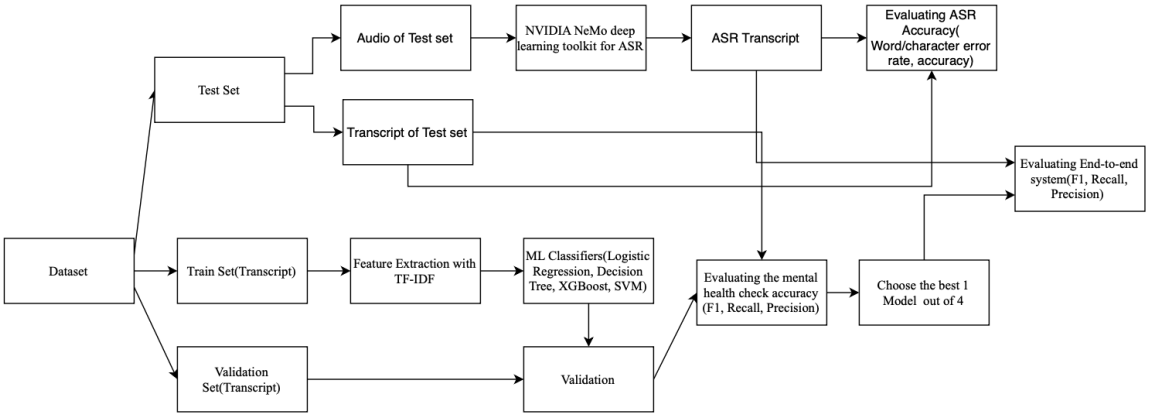


Figure 3.4. System Overview

The system begins with the DAIC-WOZ dataset, which undergoes careful preprocessing including audio enhancement, temporal alignment, and participant speech extraction. The test set is then transcribed using the NVIDIA Nemo ASR system, with transcription quality evaluated through standard metrics including Word Error Rate (WER) and Character Error Rate (CER). For mental health classification, the entire dataset's transcripts are processed using TF-IDF for feature extraction, followed by multiple machine learning classifiers including Decision Trees, Logistic Regression, and XGBoost. The best-performing model is selected based on comprehensive evaluation metrics, and finally, the ASR-transcribed test set is evaluated against this optimal model to assess the real-world performance of the complete automated assessment pipeline.

3.3 Automatic Speech Recognition Module

3.3.1 Conformer and CTC Architectures

Automatic Speech Recognition (ASR) systems have evolved toward end-to-end architectures that directly map speech to text. This work utilizes a Conformer-CTC model, which combines two key innovations. First is the Conformer architecture for feature extraction and Connectionist Temporal Classification (CTC) for sequence alignment.

The Conformer (Convolution-augmented Transformer) architecture was developed to overcome the limitations of pure convolutional or attention-based models [40]. It combines convolutional neural networks with multi-head self-attention mechanisms. This hybrid architecture enables the capture of both local acoustic patterns and global contextual dependencies at the same time, which is a critical capability for processing conversational speech. As demonstrated in [40], the Conformer architecture achieves state-of-the-art performance in automatic speech recognition by combining convolution with attention to model both local and global dependencies effectively.

Connectionist Temporal Classification (CTC) is a training method that does not require time-aligned transcripts [40]. Unlike autoregressive models that generate text word by word, CTC processes the entire audio sequence in parallel. This makes it efficient for longer recordings and robust to the pauses, corrections, and emotional variations often found in clinical interviews.

3.3.2 NVIDIA NeMo Conformer-CTC Large

This work employs the NVIDIA NeMo Conformer-CTC Large pre-trained model (stt_en_conformer_ctc_large, version 1.10.0). According to official documentation, this large-scale implementation contains approximately 120 million parameters and was trained on the NeMo ASRSET 3.0 corpus [41].

The model follows the standard Conformer-CTC architecture described in [40] and utilizes a SentencePiece Unigram tokenizer with a compact 128-token vocabulary, producing lowercase English text output. Benchmark performance on LibriSpeech datasets demonstrates the model's strong generalization capabilities: it achieves a Word Error Rate (WER) of 4.5% on the challenging test-other subset and 2.1% on test-clean using greedy decoding (selecting the most probable token at each timestep without beam search) [41] . These results indicate robust performance on diverse speech recognition tasks before any domain-specific adaptation.

This specific pre-trained model was selected for several reasons. Mainly for its proven accuracy on standard benchmarks, efficient non-autoregressive CTC inference suitable for processing lengthy clinical interviews, comprehensive training on thousands of hours of diverse English speech, and accessibility through the open-source NeMo toolkit. The model's architecture, illustrating the integration of convolutional and attention-based components, is shown in Figure 3.5 [42]. The architecture processes audio through stacked Conformer blocks, each containing feed-forward modules, multi-head self-attention layers for capturing global context, and convolutional modules for extracting local acoustic features. This hybrid design enables simultaneous modeling of fine-grained phonetic details and broader conversational structure.

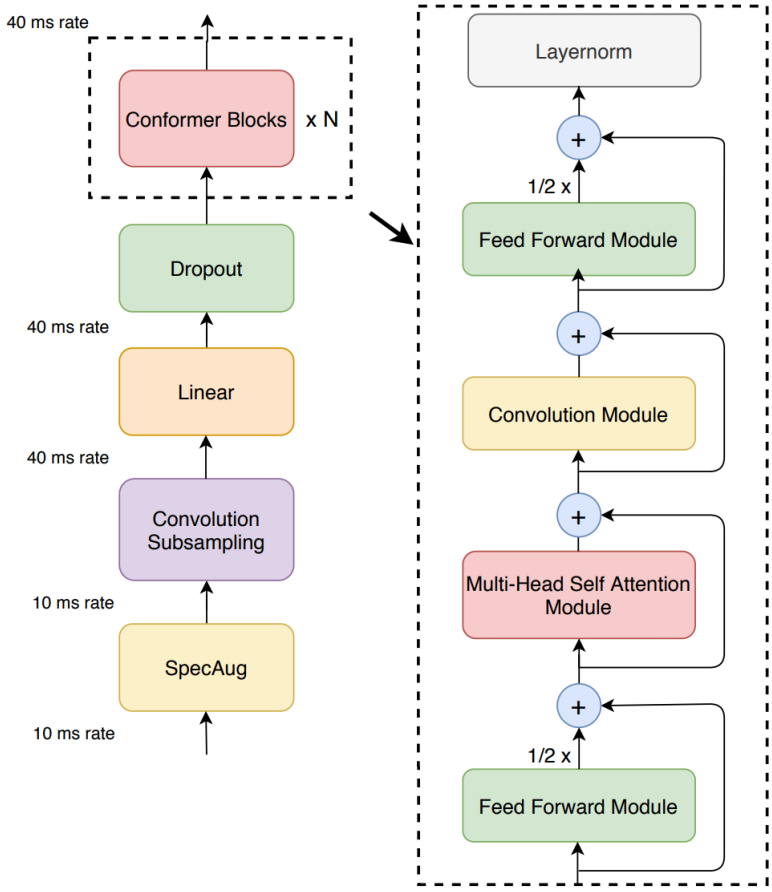


Figure 3.5. Structure of Conformer encoder [42]

3.3.3 Implementation of the toolkit

Applying a general-purpose ASR model to clinical speech presents a significant domain adaptation challenge. To ensure systematic development and reproducibility, this work implements the ASR pipeline using the NVIDIA NeMo deep learning toolkit [41]. Clinical interviews in the DAIC-WOZ corpus differ from the model's training data in several key aspects, including emotional prosody, specialized clinical terminology, and spontaneous speech patterns with hesitations and self-corrections.

This study applies the model without any clinical fine-tuning, assessing its practical suitability for real-world mental health screening. This methodological choice serves two purposes. First, it evaluates whether current general-purpose ASR technology is sufficiently accurate for immediate clinical application without requiring costly domain-specific training data or computational resources. Second, it establishes a clear performance baseline that can inform future domain-adapted models and quantify the potential gains from clinical fine-tuning.

The model processes the preprocessed audio files from the test set (as described in Section 3.1) through stacked Conformer encoder blocks, with the CTC decoder generating the most probable text sequence. The model's output is lowercase text. Participant speech was then isolated by filtering out the virtual interviewer's dialogue using speaker labels from the original transcripts. This processed output was formatted to enable direct comparison with ground-truth manual transcripts for subsequent error analysis and mental health classification. The transcription errors analyzed in Chapter 4 specifically identify which aspects of clinical speech pose the greatest challenges to current ASR systems.

3.4 Feature Representation for Speech-Based Mental Health Assessment

This section details the methodology for converting clinical speech into a numerical format suitable for machine learning. The design follows from an investigation of the fundamental trade-offs in feature representation for audio-based classification. The primary alternatives are extracting acoustic features directly from the audio signal or extracting linguistic features from transcribed text. Acoustic features, such as MFCCs, pitch, and jitter, model how speech is produced. Linguistic features, derived from text, capture what is said. As established in the Related Work Chapter, empirical evidence from multimodal studies demonstrates that linguistic features contribute more to diagnostic accuracy than acoustic features for mental health assessment, with acoustics often providing only marginal gains [27, 28]. This evidence forms the core conceptual framework for the model design. For a classification task based on semantic content, linguistic features derived from text provide a more effective representation. Consequently, the central challenge becomes obtaining accurate text from clinical audio, which leads to the implementation trade-off between manual and automated transcription.

The method applies a streamlined pipeline to address a well-known challenge in using speech technology for mental healthcare. Specifically, it tackles the inconsistent performance of Automatic Speech Recognition when transcribing clinically important speech content. Research shows that while ASR provides a scalable way to collect symptom information, its accuracy for key diagnostic terms can vary widely. Word-level recognition rates for depression-related terms range from 60% to 100% [43]. This variability introduces noise that can strongly impact the performance of later classification models. This work examines this trade-off by comparing a classification pipeline that uses high-quality manual transcripts with one that uses practical ASR-generated transcripts.

The pipeline includes two stages designed with clinical interpretability in mind. The first stage, speech-

to-text conversion, defines the main experimental variable. The manual transcripts, which serve as the highest quality reference and allow for direct inspection, are contrasted with the output of an ASR system. This reflects the clinical need for transparent data during training and supervision, an advantage noted over black box approaches like BERT or Wav2Vec2.0 [43]. The second stage, text-to-vector transformation, converts the language from either source into numerical features using the Term Frequency–Inverse Document Frequency (TF-IDF) method.

This controlled setup makes it possible to isolate transcription noise as a single factor. Manual transcripts provide a near-optimal linguistic signal, while ASR transcripts introduce the word-level errors and substitutions typical of current technology. By training and evaluating models with both input types, this study measures how well an interpretable linguistic model can tolerate imperfect automated transcription, which is expected in real-world use.

The choice of TF-IDF is driven by the need for clinical transparency and strong performance with limited data. The interpretability of text-based features is essential. TF-IDF produces features grounded in word frequencies, which supports the wider requirement for understandable clinical tools and for fair performance across different patient groups [43]. In addition, TF-IDF is a well-established statistical approach that performs stable feature selection and works effectively with traditional classifiers on smaller clinical datasets. This ensures reliable modeling with the available data, reducing the risk of overfitting that more complex, data-intensive deep learning embeddings might introduce.

3.4.1 TF-IDF Feature Extraction

This subsection describes the main feature extraction method used to convert clinical transcripts into numerical data suitable for machine learning algorithms. Term Frequency-Inverse Document Frequency (TF-IDF) is a weighting scheme that has proven its usage in the field of computational mental health research, as it identifies which words are most common and distinctive across clinical transcripts. In this study TF-IDF was successfully applied, because of its balanced calculation method, which considers how often a word occurs in single document and how common it is across all documents [32].

First component of the TF-IDF calculation is the Term Frequency (TF), it is the measure of the frequency of a word occurring in each document, since words having higher frequencies are more relevant for defining its content. The term frequency of the word i in document j is computed as:

$$TF_{i,j} = \frac{Number\ of\ times\ word\ i\ appears\ in\ document\ j}{Total\ number\ of\ words\ in\ document\ j} \tag{3.1}$$

Inverse Document Frequency (IDF) measures the uniqueness of a word across the corpus. Words that appear in many documents receive lower weights, as they offer less discriminative power. The inverse document frequency is calculated as:

$$IDF_i = ln\ (\frac{N}{df_i}) \tag{3.2}$$

where N is the total number of documents and df_i is the number of documents containing word i. The final TF-IDF weight is the product of these components:

$$TF - IDF_{i,j} = TF_{i,j} \times IDF_i \tag{3.3}$$

In this work TF-IDF was implemented using the TfidfVectorizer class from scikit-learn library in Python. The library used a smoothed version of the IDF formula to prevent division by zero:

$$IDF_i = \ln\left(\frac{N+1}{df_i+1}\right) + 1 \tag{3.4}$$

This smoothing ensures that terms appearing in all documents are not entirely ignored, while still assigning them lower importance. To understand the calculation the sample below can be discussed:
D1: “sleep is difficult most nights feel tired”
D2: “enjoy activities feel happy lately”
D3: “sleep disrupted feel exhausted all day”
The word to be explored “tired”, and it appears only in the document 1. To calculate its TF-IDF:

$$TF = \frac{1}{6} \approx 0.167$$

, because it appears once in the 6 words long document.

$$IDF = ln\left(\frac{3+1}{1+1}\right) + 1 \approx 1.693$$

, because this word appears only in one of the three documents. Then the final value:

$$TF - IDF = TF \times IDF = 0.167 \times 1.693 \approx 0.283$$

To understand this better, compare this to the word “feel”, which appears in all three documents. Its IDF calculation will be:

$$IDF = \ln\left(\frac{3+1}{3+1}\right) + 1 = 1$$

This IDF result reduces the word’s overall weight, which means this word provides less information for classification. Words like “tired” that appear in fewer documents receive higher weights and become more valuable features for the classification models.

The TF-IDF method transforms each transcript into a numerical vector, where each position represents a word from the vocabulary established during the training phase. The size of this vocabulary dictates the length of the vector. For this work, TF-IDF vectorization was configured with parameters optimized for clinical text classification (detailed in Table 3.3). The vocabulary was limited to 10,000 features to balance discriminative power with computational efficiency. Both unigrams and bigrams (n-gram range 1-2) were included to capture both individual terms and contextual word pairs. Terms appearing in fewer than 2 documents (minimum document frequency = 2) were excluded to eliminate rare terms with limited predictive value, while terms appearing in more than 80% of documents (maximum document frequency = 0.8) were filtered as overly common. This configuration ensures the feature space emphasizes terms that are sufficiently frequent to be reliable while maintaining discriminative power across the "At-Risk" and "Not At-Risk" classes. Once all transcripts are processed using this uniform approach, the resulting feature vectors are input into the machine learning classifiers, which will be elaborated upon in the following section.

3.5 Classifiers Implementation

To ensure a comprehensive evaluation, four distinct machine learning classifiers were selected. The following subsections detail the theoretical principles and implementation specifics of each selected model.

3.5.1 Decision Tree

The first classifier evaluated in this study is the Decision Tree classifier, implemented using the CART (Classification and Regression Trees) algorithm, it learns interpretable decision rules from TF-IDF vectorized text data for binary mental health classification.

Decision Trees represent a type of supervised learning model where predictions are done by partitioning data based on a hierarchical structure of decision rules. In the case of the CART algorithm, it creates a binary tree such that every internal node is split into exactly two child nodes according to a feature threshold [44]. Construction starts at a root node comprising all training samples and recursively splits the data into smaller, and homogeneous subsets. The aim is for the resulting leaf nodes to contain samples predominantly from a single class, either "At-Risk" or "Not At-Risk" regarding mental health conditions.

The CART algorithms use the Gini Impurity measure to evaluate the quality of any potential split. The Gini Impurity defines the heterogeneity of a node, it is zero when a node contains samples from only one class, and its maximum value of 0.5 when a node contains equal proportions of both classes [44]. For a binary classification problem, the Gini Index for a node T is defined as:

$$Gini\ (T) = 1 - (P_0^2 + P_1^2) \tag{3.5}$$

where P_0 is the proportion of "Not At-Risk" samples in the node and P_1 is the proportion of "At-Risk" samples.

To find the best split, the algorithm tests all available TF-IDF features and all possible threshold values. For a given feature A and a threshold t, it divides the dataset T into two subsets:
T_{left}: Samples where the TF-IDF value for feature A is ≤ t
T_{right}: Samples where the TF-IDF value for feature A is > t
The quality of this split is evaluated using the weighted Gini Index of the resulting child nodes:

$$\text{Gini(T, A)} = \frac{|T_{left}|}{|T|} Gini(T_{left}) + \frac{|T_{right}|}{|T|} Gini(T_{right}) \tag{3.6}$$

The feature and threshold pair that results in the smallest $Gini(T, A)$ is selected for the split, as this creates the greatest increase in node purity.

The practical application of these formulas occurs exclusively during the model training phase. The algorithm iterates through the entire feature space, calculating the potential Gini impurity for every possible split (feature and threshold combination) according to Equation (3.6). It selects and stores the split that yields the minimum Gini (T, A) for each node. Once training is complete, the final model is a static hierarchy of these pre-computed decision rules. During inference (classifying a new transcript), the model simply traverses this fixed tree by comparing the transcript's TF-IDF values against the stored thresholds, as illustrated in the following example. No impurity calculations are performed at this stage.

Mental health classification using the CART decision tree algorithm proceeds through a hierarchical series of binary splits based on TF-IDF features. This process can be illustrated with a simplified example. Consider a participant's text: "Lately, I have been feeling so worthless. I just want to cry all day". During the model training phase, all transcripts from the clinical corpus are collectively transformed into a TF-IDF matrix. This process assigns a weight to each term that reflects its importance in a single document relative to its commonness across all documents. Therefore, in the trained model, a term like 'worthless' is considered important not merely for its presence in this

example, but because it is a relatively rare and distinctive marker within the broader corpus of participant speech. The trained decision tree evaluates this example's TF-IDF vector by traversing its learned rule structure. It might first check if the TF-IDF weight for 'worthless' exceeds a specific learned threshold, a value optimized during training. If the weight is sufficiently high, the path proceeds to a subsequent node. This next node could then evaluate another discriminative feature, such as the TF-IDF weight for 'cry.' If this condition is also met, the traversal continues down the branch, eventually reaching a leaf node that yields the final classification of "At Risk." This structured process demonstrates how the model maps combinations of significant linguistic features to a clinical prediction via a transparent, rule-based pathway, providing an interpretable rationale for each decision.

While Decision Trees offer valuable interpretability, they have inherent limitations that must be managed through careful hyperparameter tuning. Trees can overfit training data by creating overly complex rules that capture noise rather than true patterns, particularly when allowed to grow without constraints. This is why parameters such as maximum tree depth, minimum samples per split, and minimum samples per leaf are critical. They prevent the model from memorizing training examples and instead encourage generalization to new data. The specific hyperparameters for this model, including the maximum tree depth and the criterion for splitting, were determined through a systematic grid search and cross-validation process, which is detailed in Section 3.6 .

3.5.2 Logistic Regression

The second classifier evaluated in this study is Logistic Regression. It is a widely used statistical model for binary classification, which offers strong predictive performance along with interpretable results from the model parameters. Compared to other models, Logistic Regression models the relationship between features and class probabilities using a linear combination followed by a logistic function.

For this binary mental health classification using the DAIC-WOZ transcripts, the model estimates the probability that a participant's transcript belongs to the "At-Risk" class based on TF-IDF vectorized features. The fundamental concept involves modeling the log-odds of the positive class as a linear function of the input features. Given a transcript's TF-IDF vector $t_j = [t_{j_1}, t_{j_2}, \ldots, t_{j_n},]$, the model computes [45]:

$$log(\frac{P(C = 1|t_j)}{1 - P(C = 1|t_j}) = \beta_0 + \beta_1 t_{j_1} + \beta_2 t_{j_2} + \ldots + \beta_n t_{j_n} \tag{3.7}$$

where $P(C = 1|t_j)$ represents the probability that the transcript belongs to the "At-Risk" class, β_0 is the intercept term, and $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients corresponding to each TF-IDF feature.

The probability itself is obtained through the logistic sigmoid function:

$$P(C = 1|t_j) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 t_{j_1} + \beta_2 t_{j_2} + \ldots + \beta_n t_{j_n})}} \tag{3.8}$$

This function constrains the output to a value between 0 and 1, representing a valid probability. The model parameters $\beta_0, \beta_1, \ldots, \beta_n$ are estimated through maximum likelihood estimation (MLE), which finds the parameter values that make the observed training data most probable. This is achieved by minimizing the cross-entropy loss function with L2 regularization to prevent overfitting:

$$L(\beta) = -\sum_{j=1}^{m} [y_j log\,(P(C = 1|t_j)) + (1 - y_j) log(1 - P(C = 1|t_j))] + \lambda \sum_{k=1}^{n} \beta_k^2 \tag{3.9}$$

where y_j is the true label of transcript j, m is the number of training samples, and λ controls strength of regularization to prevent overfitting by correcting large coefficient values.

The classification process can be illustrated with the same participant transcript: “Lately, I have been feeling so worthless. I just want to cry all day”. The model evaluates this text by computing a weighted sum of its TF-IDF features, where each feature is multiplied by its learned coefficient. Words like 'worthless' and 'cry' that strongly indicate mental health risk receive positive coefficients during training. When such terms appear with substantial TF-IDF weights in a transcript, they contribute to a higher log-odds value, which the sigmoid function transforms into a probability of being 'At-Risk.' If this probability exceeds the threshold of 0.5, the transcript is classified accordingly. The model's reliance on explicit, weighted combinations of linguistic features provides clinicians with clear insight into which specific words and phrases drove the classification decision.

The trained model provides direct clinical interpretability through its learned coefficients. Positive coefficients indicate words that increase the likelihood of an 'At-Risk' classification, while negative coefficients correspond to words associated with the 'Not At-Risk' class. For classifying new transcripts, the model calculates the probability score using Equation (3.8) and classifies a transcript as 'At-Risk' if this probability exceeds the 0.5 threshold, providing both a classification decision and a confidence measure. This transparency allows clinicians to understand not only the outcome but also the specific linguistic signals the model relies upon. The strength of the L2 regularization parameter $\mathrm{C} = 1/\lambda$ was optimized through the grid search process.

3.5.3 XGBoost

XGBoost, which stands for Extreme Gradient Boosting, is a scalable and highly efficient machine learning algorithm based on the gradient boosting framework [46]. Its core principle is to combine a sequence of simple weak learners. In this case, the weak learners are decision trees. The goal is to create a single strong predictive model. Each new tree added to the sequence is trained to correct the residual errors, or mistakes, made by the current group of trees.

The model's final prediction for an instance result from adding together the outputs of all K trees:

$$\widehat{y_i} = \sum_{k=1}^{K} f_k(x_i) \tag{3.10}$$

A key innovation of XGBoost is its sophisticated objective function. This function is designed to be optimized every time a new tree is added. It consists of two critical parts. The first part is a training loss term, $L(\theta)$, which quantifies how well the model fits the training data. The second part is a regularization term, noted as $\Omega(\theta)$, which controls model complexity to prevent overfitting [47]. This combined objective is defined as follows:

$$\mathrm{Obj}(\theta) = \mathrm{L}(\theta) + \Omega(\theta) \tag{3.11}$$

For a single tree model f, the regularization term is defined as follows.

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{3.12}$$

where T is the number of leaves in the tree, w_j is the score on each leaf, and γ and λ are regularization parameters that control the penalty for the number of leaves and the magnitude of the leaf scores, respectively.

XGBoost optimizes this objective function efficiently by using a second order Taylor approximation. This method considers both the gradient (the first derivative) and the Hessian (the second derivative) of the loss function. This leads to faster and more accurate convergence compared to traditional gradient boosting. The algorithm constructs the model in a greedy manner. For every possible split during tree building, it calculates a gain value. This gain measures whether the reduction in prediction error from the split is greater than the cost of added complexity from the regularization term γ. If the gain is not sufficient, the proposed branch is pruned and not added to the tree. This built in regularization is the primary advantage of XGBoost over a standard Decision Tree, as it creates a robust ensemble of trees that generalizes much more effectively to new, unseen data [47].

In this work, the XGBoost model was applied to TF IDF vectorized transcripts. The ensemble builds a sequence of decision trees. The first tree may make predictions based on the most obvious terms, such as those with high TF IDF weights linked to negative emotion. Each tree added after the first is then trained on the residual errors from the current ensemble. This allows the model to focus iteratively on cases that were previously misclassified or on subtler linguistic patterns. For example, one tree might identify strong negative words like "worthless". A subsequent tree could learn to recognize transcripts that lack positive language about the future. Another tree might detect patterns related to social isolation. The final prediction is the sum of the outputs from all trees in the ensemble. This process integrates a wide range of linguistic signals for a more robust assessment. The algorithm's inherent regularization continuously prunes non-essential branches, ensuring a model that balances performance with generalization. The specific hyperparameters for this model, including the learning rate, maximum tree depth, and the number of trees, were optimized through the systematic grid search and cross validation process detailed in Section 3.6 .

3.5.4 Support Vector Machine

Support Vector Machine (SVM) is one of the most popular supervised learning approaches due to its strong theoretical foundation and effectiveness in high-dimensional classification problems. The key concept behind the SVM is to find the maximum-margin separating hyperplane between two classes in the feature space. For mental health classification using DAIC-WOZ transcripts, the SVM distinguishes between “At-Risk” and “Not At-Risk” categories within the high-dimensional space constructed by TF-IDF vectorization.

In this work, a Linear SVM was employed to identify an optimal linear decision boundary described by the function [48]:

$$f(x) = w * x + b \tag{3.13}$$

where w is the learned weight vector, x represents the TF-IDF feature vector of a transcript, and b is the bias term. The classification is determined by the sign of this function: if $f(x) > 0$, the transcript is classified as “At-Risk”, otherwise it is classified as “Not At-Risk”.

The optimal hyperplane is derived by solving a constrained optimization problem that maximizes the margin between classes while minimizing training error. This optimization, formulated with L2 regularization, is expressed as:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} max(0,1 - y_i(w * x_i + b)) \tag{3.14}$$

where C is the penalty parameter that controls the trade-off between margin maximization and training error minimization, $y_i \in \{-1, +1\}$ is the label of training sample I, and $x_i \in \mathbb{R}^m$ is its corresponding

TF-IDF vector. The term $max(0,1 - y_i(w * x_i + b))$ represents the hinge loss, which penalizes misclassifications and points that fall within the margin. A higher C value prioritizes fitting the training data closely, while a lower C promotes better generalization.

Consider the same example transcript, “Lately, I have been feeling so worthless. I just want to cry all day”. Its TF-IDF vector x is input into Equation (3.13). The model computes the dot product $w * x$, where w contains learned weights. Words like "worthless”, and "cry" will have strong positive weights in w. Their high TF-IDF values in this transcript result in a large positive contribution to the sum $w * x$. The model then adds the learned bias term b, which adjusts for the overall class distribution in the training data. If the final value $f(x) = w * x + b$ is positive, the transcript is classified as "At-Risk". The magnitude of f(x) indicates the model's confidence, reflecting how distinctly the linguistic pattern aligns with the "At-Risk" side of the learned hyperplane.

3.6 Model Training and Testing Procedure

This section details the comprehensive experimental framework employed to develop and validate the mental health classification system. The methodology was designed to systematically evaluate both the feature extraction capabilities of TF-IDF vectorization and classification performance of multiple machine learning algorithms, while maintaining exact experimental standards suitable for clinical applications.

The experimental design was based on structured two-phase model for a baseline analysis and system viability checks. Phase 1 aimed at constructing the best classifiers with manually transcribed interviews from the official dataset, whereas in phase 2 degradation in audio performance when applying ASR transcripts under same conditions was assessed. The dataset underwent careful partitioning using participant level split. to prevent data leakage. The full dataset was divided into training (71%, 131 participants), validation (13%, 23 participants), and test sets (15%, 28 participants), the split of the sets with participant numbers can be seen in Table 3.2. This approach ensured the consistency between the three sets, preserving natural correlation between utterances and providing realistic performance estimates.

Excluded Participants	
342, 394, 398, 460, 373, 444, 451, 458, 480, 402	
Training Set	
300, 301, 303, 304, 305, 306, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 395, 396, 397, 399, 400, 401, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 438, 439	
Validation Set	**Test set**
302, 307, 331, 472, 473, 474, 475, 476, 477, 478, 479, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492	440, 441, 442, 443, 445, 446, 447, 448, 449, 450, 452, 453, 454, 455, 456, 457, 459, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471

Table 3.2. Data Split by Participant Number

3.6.1 Class Imbalance Handling

As illustrated in Figure 3.2, the dataset exhibits significant class imbalance, with approximately twice as many as “Not At-Risk” class participants (PHQ-8<10) compared to “At-Risk” cases. This 2:1 ratio reflects real world clinical prevalence but causes challenges for machine learning model training,

potentially causing bias toward the majority class and compromised minority class detection performance.

Research indicates that oversampling techniques like SMOTE and ROS exhibit varying sensitivity to different classifier algorithms, where no single resampling method consistently optimizes performance across all models [49]. Based on these findings, class weighting at the algorithm level was selected as the primary strategy to address imbalance while ensuring consistent performance across diverse classifiers.

Class weighting addresses imbalance by assigning different costs to misclassifying samples from different classes during model training. The weight for each class was calculated using the balanced class weighting formula:

$$w_c = \frac{n_{samples}}{n_{classes} \times n_{samples_c}} \tag{3.15}$$

where w_c is the wight for class c, $n_{samples}$ is the total number of samples, $n_{classes}$ is the number of classes, which is 2 in this case, and $n_{samples_c}$ is the number of samples in class c.

For this study, the training set consisted of 87 "Not At-Risk" and 44 "At-Risk" participants, resulting in the following calculated weights:

- "Not At-Risk" weight: $w_0 = \frac{131}{2\times87}$
- "At-Risk "weight:$w_1 = \frac{131}{2\times44}$

This means misclassifying an "At-Risk" participant carries approximately twice the penalty compared to misclassifying a "Not At-Risk" participant.

The class weighting strategy was applied consistently across all machine learning algorithms using the class_weight='balanced' parameter in scikit-learn implementations. All models were trained on the original, imbalanced training data while using these calculated weights to adjust their loss functions during optimization. Validation and test sets maintained their natural class distributions throughout evaluation.

3.6.2 TF-IDF Configuration for Model Training

For feature extraction the TF-IDF vectorization parameters were selected of clinical text classification, details in Table 3.3. The implementation employed TfidfVectorizer class from scikit-learn library, and the parameters were chosen to balance feature discriminability and computational efficiency for this exact task.

Parameter	Configuration	Explanation
Maximum Features	10000	Balance dimensionality and computational efficiency
N-gram Range	(1,2)	Capture both unigrams and contextual bigrams
Minimum Document Frequency	2	Eliminate rare terms with limited predictive power
Maximum Document Frequency	0.8	Remove overly common terms with low discriminative value
Stop Words	English	Reduce noise from common function words
Smooth IDF	Enabled	Prevent division by zero for unseen terms
Norm	L2	Normalize feature vectors for consistent scaling

Table 3.3. TF-IDF Feature Extraction Parameters

The vectorizer with consistent transformation was fitted only on training data to prevent information leakage, and then it was applied to validation and test sets using the same feature space. This approach ensured that models learned from the training data without exposure to test set characteristics.

3.6.3 Classifier Development and Systematic Optimization

To systematically develop and validate multiple depression classification models, four distinct machine learning algorithms were implemented and fine-tuned. The first is a Support Vector Machine (SVM) with a linear kernel, which maximizes the margin and is well-suited for high-dimensional, sparse TF-IDF features. The second model is Logistic Regression, used as a baseline probabilistic classifier with L2 regularization. The third is a Decision Tree based on the CART algorithm and using the Gini Impurity criterion. Finally, XGBoost is included as a gradient boosting method with built-in regularization, allowing it to capture more complex feature interactions in clinical language patterns.

A comprehensive hyperparameter tuning process was implemented using Grid Search with 3-fold stratified cross-validation, optimizing for weighted F1-score while maintaining participant-level data integrity. This approach aligns with established methodologies in clinical prediction research, where Grid Search optimization has demonstrated statistically significant improvements in model accuracy for healthcare applications [50]. The parameter search spaces, and optimal selections are detailed in Table 3.4.

Classifier	Parameters	Search Values	Optimal Selection
Linear SVM	Regularization (C)	[0.01, 0.1, 1.0]	0.1
	Max Iterations	[1000, 5000]	5000
Logistic Regression	Regularization (C)	[0.01, 0.1, 1.0, 10]	1.0
	Max Iterations	[1000, 5000]	5000
Decision Tree	Maximum Depth	[10, 15, 20, None]	15
	Min Samples Split	[5, 10, 15]	10
	Criterion	['gini', 'entropy']	gini
XGBoost	Learning Rate	[0.05, 0.1, 0.2]	0.1
	Maximum Depth	[4, 6, 8]	6
	Tree Count	[200, 300, 500]	300

Table 3.4. Hyperparameter Optimization Framework

Optimization was performed with fixed random seed initialization (seed=42) to ensure reproducibility. Only the training dataset was used for hyperparameter tuning, with performance evaluation on the

validation set to prevent information leakage.

A validation strategy was implemented using three sequential layers to ensure reliable model development. First, 3-fold stratified cross-validation was applied within the Grid Search procedure to optimize hyperparameters in a systematic and unbiased manner. This was allowed by a second layer using 5-fold stratified cross-validation on the full training set to assess the stability and consistency of model performance across different data splits. Finally, the third layer involved evaluating each model on a held-out validation set, which served as the basis for final model selection. This multi-layer strategy reduces the risk of overfitting and aligns with best practices in clinical machine learning, where comprehensive validation combined with tuned hyperparameters consistently leads to more robust predictive performance [50]. Throughout all validation steps, participant-level stratification was maintained to prevent data leakage between training and validation subsets.

The complete experimental framework was implemented in Python 3.9, using scikit-learn 1.2.0 for core machine learning algorithms, imbalanced-learn 0.9.0 for class imbalance handling, and XGBoost 1.6.0 for gradient boosting implementation. All computational experiments were executed on a standardized platform with 8-core CPU architecture and 16GB RAM, establishing a systematic framework for the development and fine-tuning of multiple depression classification models.

3.7 Evaluation Metrics

This section describes the quantitative metrics used to evaluate both the Automatic Speech Recognition system and the mental health classification models. The evaluation framework employs two distinct sets of metrics aligned with the two-phase experimental design described in Section 3.6 . ASR performance metrics quantify transcription accuracy at both word and character levels, enabling assessment of how faithfully the system captures clinical speech. Classification performance metrics evaluate the mental health screening system across multiple dimensions—overall accuracy, class-specific performance, and threshold-independent discrimination—with particular emphasis on metrics that reflect clinical screening priorities such as sensitivity for detecting at-risk individuals. The choice of metrics reflects both technical performance requirements and clinical application priorities.

3.7.1 ASR Performance Metrics

The accuracy of the Automatic Speech Recognition system was evaluated using standard speech processing metrics:

Word Error Rate (WER) measures the transcription accuracy at the word level and is calculated as:

$$Word\ Error\ Rate = \frac{S + D + I}{N} \times 100\% \tag{3.16}$$

where S represents the number of substitutions (words incorrectly transcribed as different words), D the number of the deletions (words in the reference but missing from the hypothesis), I the number of insertions (extra words in the hypothesis are not present in the reference), and N the total number of words in the reference transcript. The lower word error rate values indicate more accurate transcription.

Character Error Rate (CER) provides granular assessment at the character level:

$$Character\ Error\ Rate = \frac{S_c + D_c + I_c}{N_c} \times 100\% \tag{3.17}$$

where S_c, D_c, I_c represent character level substitutions, deletions and insertions respectively, and N_c

is the total number of characters in the reference transcript. Character error rate is particularly valuable for evaluating partial word recognition and assessing whether transcription errors preserve semantic content through recognizable partial forms.

ASR Accuracy quantifies overall transcription quality by measuring the proportion of correctly recognized words:

$$ASR\ Accuracy = (1 - WER) \times 100\% \tag{3.18}$$

This metric provides an intuitive measure of transcription success, with higher values indicating better performance. Together, these three metrics enable comprehensive assessment of ASR system performance on clinical interview audio, with word and character error rates identifying specific error patterns and ASR Accuracy providing an overall quality measure.

3.7.2 Classification Performance Metrics

A comprehensive set of metrics was used to evaluate the mental health classification system. These metrics provide a detailed breakdown of classification results based on the confusion matrix shown in Table 3.5, which cross-tabulates predicted classifications against actual clinical labels.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 3.5. Confusion Matrix

Accuracy measures the overall correctness of the classification system: across both classes

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.19}$$

While accuracy provides a general performance measure, it can be misleading for imbalanced datasets where high accuracy may be achieved by simply predicting the majority class. Therefore, additional class-specific metrics are essential for comprehensive evaluation.

Precision quantifies the reliability of positive predictions -the proportion of predicted "At-Risk" cases that are truly “At-Risk”:

$$Precision = \frac{TP}{TP + FP} \tag{3.20}$$

In clinical settings, precision reflects the system's ability to minimize false alarms. While false positives create additional assessment burden by flagging individuals who do not require intervention, they are generally considered acceptable in preliminary screening contexts where the primary goal is to ensure no “At-Risk” individuals are missed.

Recall measures the system’s ability to identify all actual positive cases:

$$Recall = \frac{TP}{TP + FN} \tag{3.21}$$

In mental health screening applications, recall is the most clinically critical metric. It quantifies the proportion of truly “At-Risk” individuals correctly identified by the system. Failing to identify an individual requiring intervention (false negative) carries far greater clinical consequences than

producing a false alarm (false positive), as missed cases represent lost opportunities for potentially life-saving intervention.

F1-Score provides the harmonic mean of precision and recall, offering a balanced metric particularly valuable for imbalanced datasets:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3.22}$$

Area Under the ROC Curve (AUC-ROC) evaluates the model's ability to distinguish between classes across all possible classification thresholds. The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (Recall) against the False Positive Rate at various decision thresholds. The area under this curve provides a threshold-independent performance measure, with values ranging from 0.5 to 1.0. AUC-ROC is particularly valuable for comparing different classifiers, as it captures their discrimination capability independent of any specific operating threshold. ROC curves for each classifier are presented in Chapter 4, enabling direct visual comparison of classifier performance and threshold selection strategies.

Chapter 4

4 Experimental Results and Performance Evaluation

4.1 Performance of the Automatic Speech Recognition (ASR) System

This section assesses the effectiveness of the NVIDIA NeMo Toolkit Encoder CTC Large model in transcribing clinical interview data, focusing on both quantitative accuracy metrics and the qualitative findings of transcripts characteristics.

4.1.1 Overall ASR Performance on Clinical Data

The ASR system exhibited strong performance across 28 clinical interviews, effectively handling spontaneous therapeutic dialogue with an average transcription accuracy of 86.3%. It achieved a mean word error rate (WER) of 13.2% and a character error rate (CER) of 7.4%.

Performance Metric	Mean	Standard Deviation	Range	25th Percentile	75th Percentile
Accuracy (%)	86.3	5.8	70.2-93.5	83.3	89.7
Word Error Rate (%)	13.2	5.8	6.4-29.8	9.7	15.5
Character Error Rate (%)	7.4	3.9	3.4-23.1	5.0	8.0
Interview Duration (s)	427.6	159.2	175.1-868.2	321.1	544.4

Table 4.1. Comprehensive ASR Performance Metrics

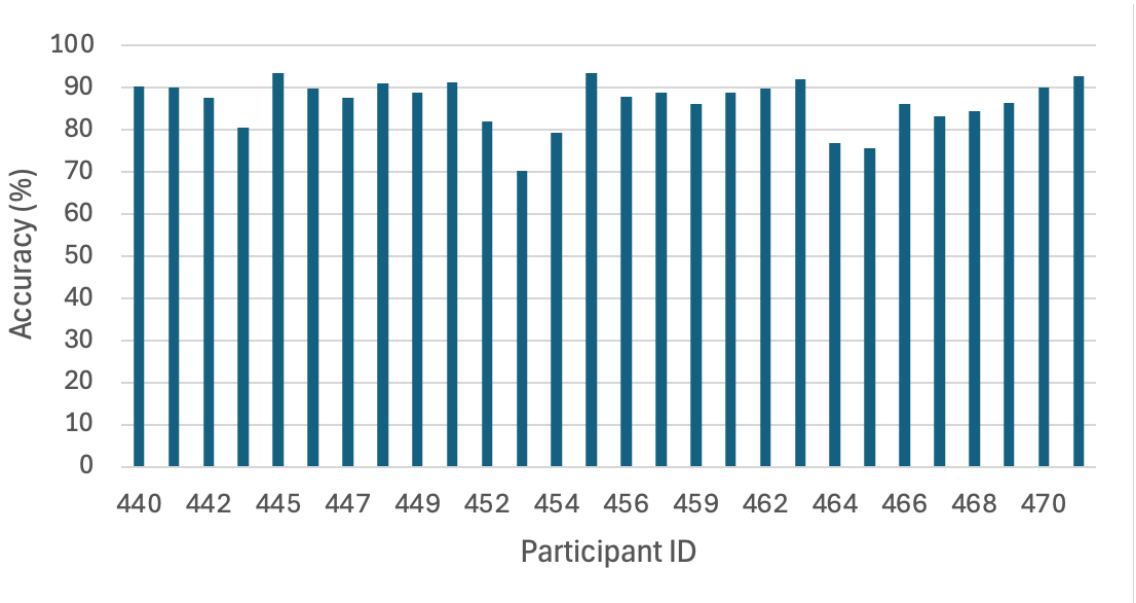


Figure 4.1.Distribution of ASR Accuracy Scores

The distribution of performance indicates that 85.7% of participants (24/28) attained accuracy scores

above 80%, and almost one-third (28.6%) surpassed the excellence threshold with accuracy rates over 90%. The character error rate remained consistently lower than the word error rate across all participants, indicating that while complete word errors occurred, the system more frequently produced partial word forms or minor spelling variations that preserve semantic content.

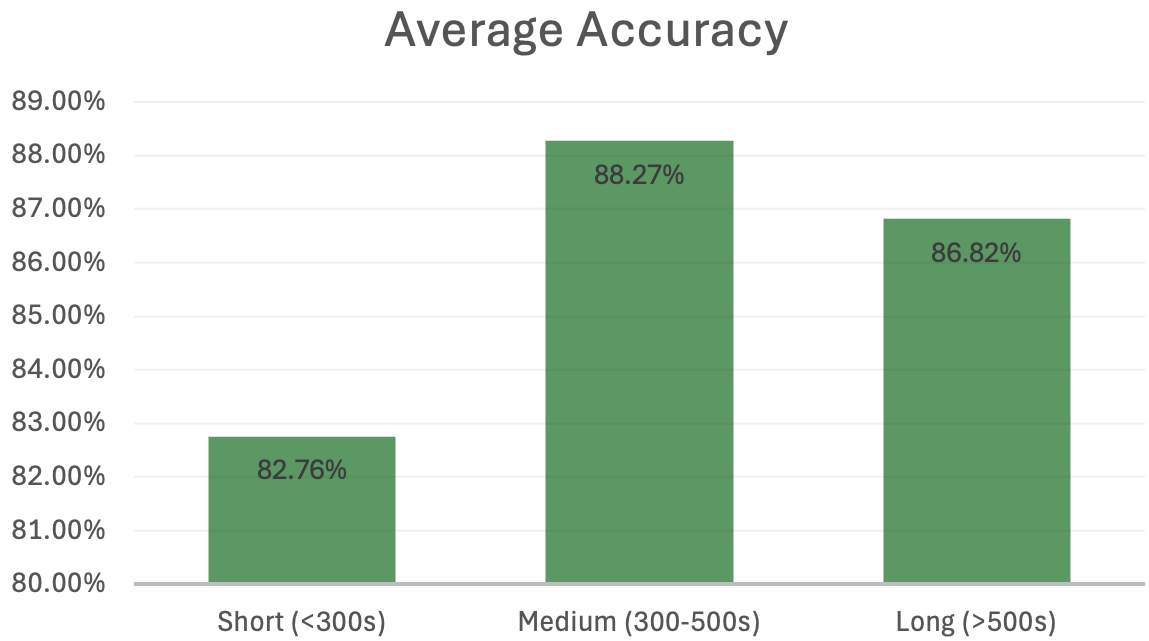


Figure 4.2.Performance Stability Across Interview Durations

An analysis of interview duration revealed stable performance across different session lengths. Medium-length interviews, lasting between 300 and 500 seconds, achieved the highest accuracy at 88.27%. Shorter sessions, those under 300 seconds, exhibited slightly lower performance, with an accuracy rate of 82.76%. Meanwhile, longer clinical sessions, exceeding 500 seconds, maintained a strong accuracy of 86.82%.

4.1.2 Qualitative Analysis of ASR Transcripts

To understand how transcription quality varies across clinical presentations, two cases representing the performance range were analyzed: a high performing case (93.5% accuracy, Participant 445) and a lower performing case (70.2% accuracy, Participant 453).

Participant 453 exemplifies challenges encountered in clinical speech recognition through speech patterns typical of therapeutic settings: rapid pacing, emotional inflection, and frequent self-corrections. Medical terminology related to psychological conditions proved particularly vulnerable to errors, with terms like “PTSD” often fragmented or misrecognized. Therapeutic process descriptions and treatment-related content showed reduced accuracy. Variations in vocal volume, background acoustics, and emotional speech characteristics contributed to recognition errors.

A comparative analysis of these two cases across key clinical dimensions reveals the direct impact of these error patterns, as summarized in Table 4.2 and Table 4.3 .

Type of the Example	Original Transcript	ASR Output	Quality Assessment
Emotional State	“sometimes when things don't go my way I get depressed or upset”	“sometimes when things don't go my way i get depressed or upset”	Perfect preservation of emotional content
Trauma Narrative	“when I was taking care of my mother and she was dying of pulmonary disease”	“when i was taking care of my mother and she was dying of pulmonary disease”	Medically recognizable despite formatting differences
Therapeutic Progress	“I was released from therapy the therapist released me”	“i was released from therapy the therapist released me”	Critical treatment information fully maintained
Disfluency Handling	No prominent disfluencies in the transcript	N/A	N/A

Table 4.2.Illustrative Examples of ASR Performance of Participant 445(93.5% Accuracy)

Type of the Example	Original Transcript	ASR Output	Quality Assessment
Emotional State	“it's just the emotion it's not real”	“it's just their motion it's not real”	Semantic distortion. The core psychological term is lost, creating nonsense.
Trauma Narrative	“the p_t_s_d”	“the p t s d”	Critical fragmentation. The clinical acronym is destroyed, losing its diagnostic meaning.
Therapeutic Progress	“i was released from therapy”	“i was released from therapy”	Accurate preservation. Same key phrase is correctly captured.
Disfluency Handling	“i i was a professional athlete”	“i was a professional athlete i one time”	Error introduced. Natural repetition leads to an ungrammatical insertion.

Table 4.3. Illustrative Examples of ASR Performance of Participant 453(70.2% Accuracy)

Despite transcription errors, the ASR system preserved core clinical content across both high and lower performing cases. In high performance instances, nearly all critical mental health indicators were retained. Even in challenging cases with 70.2% accuracy, the system maintained semantic integrity sufficient for analysis. The NeMo model demonstrated strength in capturing explicit emotional vocabulary and structured narratives, though it encountered difficulties with fragmented speech patterns and specialized medical terminology.

The mean ASR accuracy of 86.3% and consistent preservation of diagnostically relevant content across the 28 test participants establishes the transcription quality baseline for this study. Section 4.3 evaluates whether this level of transcription fidelity maintains adequate classification performance for clinical mental health assessment applications.

4.2 Performance of Mental Health Classification with Manual Transcripts

This section establishes baseline classification performance using manually transcribed clinical interviews. Four machine learning algorithms were systematically compared to identify the optimal model for depression detection.

4.2.1 Overall Classifier Performance Comparison

Four machine learning algorithms (Logistic Regression, Support Vector Machine, XGBoost, and Decision Tree) were trained on 131 manually transcribed clinical interviews, from the DAIC-WOZ training set and for validat6ion 23 participants were used. As was described in Section 3.5 , each model utilized the TF-IDF feature extraction to transform transcripts into numerical representations suitable for binary classification of depression status based on PHQ-8 scores, using the threshold of 10.

Each classifier was fine-tuned using the Grid search with 3-fold stratified cross-validation on the training data after using the hyperparameter optimization procedure outlined in Section 3.6 . The optimal configurations identified through process was then evaluated on the independent test set of 28 participants.

The test set had a natural 22:6 class imbalance (78.6 % "Not At-Risk", 21.4% "At-Risk"), reflecting real clinical prevalence rates. This imbalance was maintained for realistic evaluation and used multiple metrics to assess performance. Multiple performance metrics beyond accuracy were employed to assess model behavior across both classes. The At-Risk class (PHQ-8 ≥ 10) was designated as the positive class to align with clinical screening priorities.

Classifier	Accuracy (%)	F1-Score	"At-Risk" Precision	"At-Risk "Recall
Support Vector Machine	75.0	0.76	0.45	0.83
XGBoost	75.0	0.76	0.45	0.83
Decision Tree	60.7	0.64	0.35	1.00
Logistic Regression	35.7	0.33	0.25	1.00

Table 4.4. Overall Performance Comparison of Classification Algorithms on Manual Transcripts

Support Vector Machine and XGBoost achieved identical optimal performance with 75% accuracy and 0.76 weighted F1-score. Both classifiers demonstrated balanced performance across classes: 83.3% recall for "At-Risk" detection (identifying 5 of 6 depression cases) and 72.7% recall for "Not At-Risk" participants (correctly identifying 16 of 22 healthy individuals).

Decision Tree achieved 60.7% accuracy with a distinct performance profile: 100% recall for "At-Risk" cases (detecting all 6 depression cases) but only 50% recall for "Not At-Risk" participants (11 of 22). Logistic Regression exhibited the poorest overall performance at 35.7% accuracy. Despite achieving 100% "At-Risk" recall, the model's extreme bias toward predicting depression rendered it unsuitable for practical deployment.

The performance differences between algorithms can be attributed largely to their differential responses to the class weighting strategy employed during training. Following the class balancing strategy described in Section 3.6 , all models were trained with 2:1 class weighting favoring "At-Risk" cases. SVM and XGBoost effectively integrated this weighting information, achieving balanced performance without over-correcting. In contrast, Decision Tree and Logistic Regression overcompensated for the imbalance, developing extreme biases toward the minority "At-Risk" class.

The identical performance of SVM and XGBoost is likely reflects the relatively small test size, where

both models made the same predictions. SVM was selected as the optimal model due to its computational efficiency, theoretical foundation, and established effectiveness in text classification tasks. As established by Thake et al. in a validation study of 790 participants, the PHQ-8 instrument itself achieves 74% sensitivity and 71% of specificity against clinical interviews (SCID) for detecting major depressive disorder [51]. The 75% accuracy achieved by SVM indicates the automated classification system performs at the level of the screening instrument providing the ground truth labels.

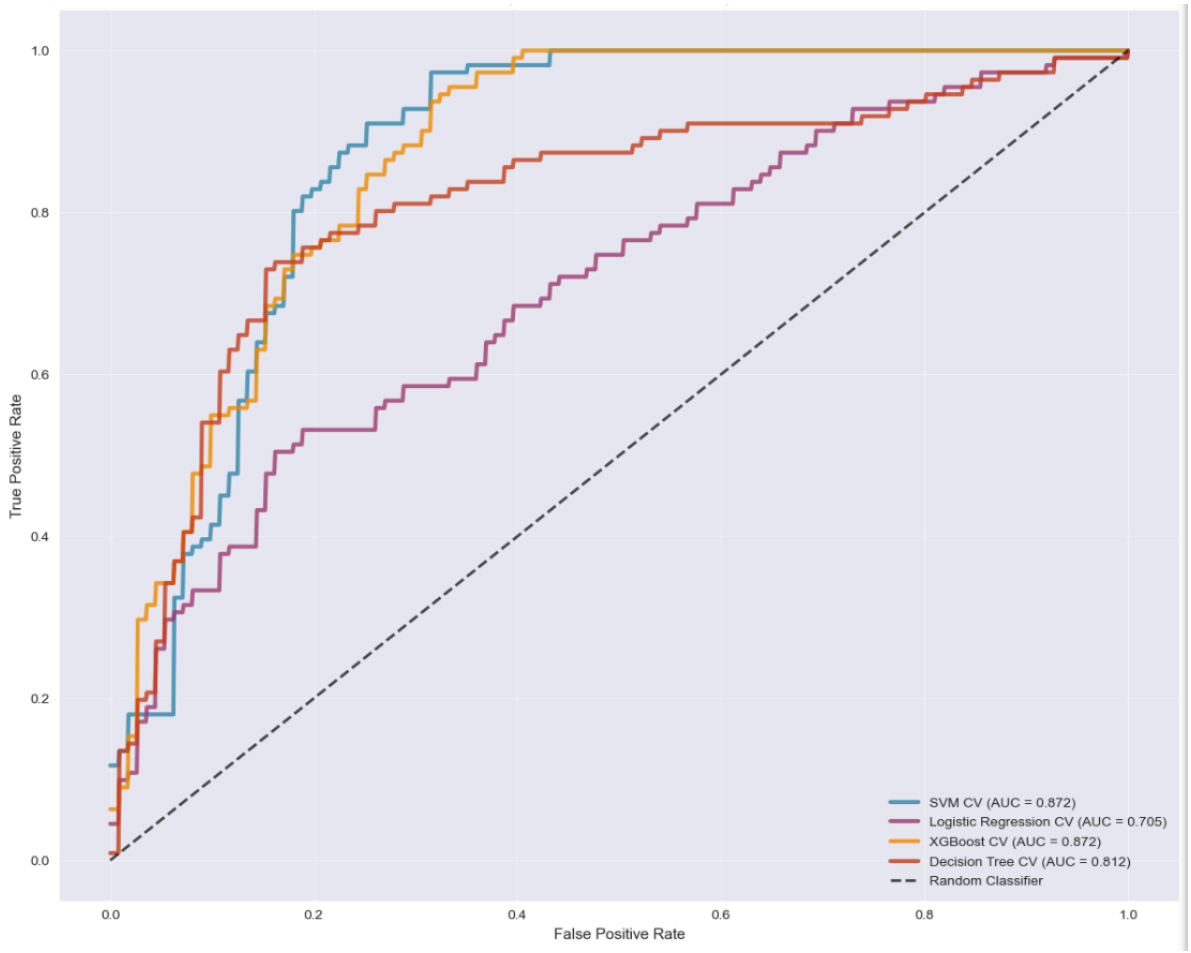


Figure 4.3.ROC Curves for All Classifiers

All four models performed substantially above baseline chance level (21.4% prevalence), validating the fundamental feasibility of TF-IDF-based depression detection from clinical interview transcripts. However, the substantial performance differences between algorithms underscore the importance of systematic model selection and hyperparameter optimization.

4.2.2 Detailed Analysis of the Optimal Classifier

This section provides the detailed analysis of the SVM classifier's performance characteristics, error patterns, and clinical implications.

	Predicted: Not At-Risk	Predicted: At-Risk
Actual: Not At-Risk	16	6
Actual: At-Risk	1	5

Table 4.5. Confusion Matrix for Support Vector Machine on Test Set (N=28)

The confusion matrix shows that the SVM classifier correctly identified 5 of 6 "At-Risk" participants

(true positives) and 16 of 22 “Not At-Risk” participants (true negatives). The model produced 1 false negative (“At-Risk” case classified as “Not At-Risk”) and 6 false positives (“Not At-Risk” cases classified as “At-Risk”).

The two types of classification errors carry different clinical implications. False negatives (1 case, 16.7% of “At-Risk” participants) represent individuals who were not identified by the system. These individuals would not receive recommended follow-up assessment. False positives (6 cases, 27.3% of Not At-Risk participants) represent individuals incorrectly flagged as potentially “At- Risk”. This reflects a conservative screening approach that prioritizes detecting “At-Risk” cases over minimizing false alarms. While false positives impose additional assessment burden on clinical workflows, they do not represent missed opportunities for intervention.

The overall accuracy of 75.0% indicates the automated system performs at the level of the screening instrument itself. The “At-Risk” recall of 83.3% demonstrates the system detects most cases exceeding the PHQ-8 threshold. “At-Risk” precision of 45.5% (5/11) indicates that slightly fewer than half of positive predictions are correct, representing substantial improvement over the 21.4% baseline prevalence in the dataset. The “Not At-Risk” recall of 72.7% shows the system correctly identifies nearly three-quarters of participants below the threshold, while “Not At-Risk” precision of 94.1% (16/17) means negative predictions are correct more than 94% of the time, providing strong confidence.

False negatives likely represent participants with subtle or atypical linguistic presentations who discuss symptoms less explicitly. The TF-IDF approach relies on discriminative term frequency; participants with mild symptoms or nuanced language may not trigger the classification threshold. False positives likely represent individuals discussing negative life circumstances, past difficulties, or emotionally laden topics without meeting the PHQ-8 threshold. The model may detect linguistic markers of general psychological distress or life stress rather than specifically the symptom patterns captured by PHQ-8 scores above 10.

4.2.3 Feature Importance and Model Interpretability

Understanding which linguistic features drive depression classification provides clinical validation of the model's decision-making process and ensures the classifier-based predictions on clinically meaningful patterns rather than spurious correlations. This section analyzes the most discriminative features identified through TF-IDF weighting in the optimal SVM classifier.

The TF-IDF vectorization process generated a feature space of 10,000 terms using unigrams and bigrams (n-gram range 1-2). Feature importance was quantified through SVM coefficient weights, which indicate each feature's contribution to the classification decision boundary. Features with larger absolute coefficient weights exert stronger influence on classification outcomes.

Feature/Word	Score	“At-Risk” mean	“Not At-Risk” mean	Class	Meaning
couldn’t	0.5	0.0276	0.0061	“At-Risk”	Functional impairment
basic	0.359	0.0381	0.0143	“At-Risk”	Self-perception
depress	0.324	0.0359	0.0138	“At-Risk”	Emotional state
cry	0.278	0.0125	0.0021	“At-Risk”	Emotional expression
manage	0.252	0.0177	0.0051	“At-Risk”	Functional capacity
medic	0.217	0.0206	0.0072	“At-Risk”	Treatment
asleep	0.206	0.0168	0.0053	“At-Risk”	Sleep disturbance
brother	0.205	0.0061	0.0219	“Not At-Risk”	Family references
son	0.181	0.0269	0.0121	“At-Risk”	Family references
anymore	0.173	0.0199	0.0079	“At-Risk”	Loss, change

Table 4.6. 10 Most Discriminative Features for Mental Health Classification

The feature importance analysis reveals clinically meaningful linguistic patterns that align with established symptomatology of mental health conditions. The most discriminative feature "couldn't" (score: 0.500) appears 4.5 times more frequently in “At-Risk” interviews, directly capturing inability to perform daily activities. This term typically appears in contexts such as "couldn't sleep," "couldn't focus," or "couldn't get out of bed," reflecting the functional impairment characteristic of mental health challenges. The related term "manage" (score: 0.252) reflects struggles with daily task execution, appearing 3.5 times more frequently in “At-Risk” cases, while "anymore" (score: 0.173) captures loss of previous abilities, reflecting the characteristic pattern where participants contrast their current diminished functioning with prior capabilities.

Direct emotional vocabulary constitutes powerful discriminators. The root term "depress" (score: 0.324) appears 2.6 times more frequently in “At-Risk” interviews, demonstrating that participants openly verbalize their emotional state using clinical terminology. The term "cry" (score: 0.278) shows the strongest frequency differential at 6.0-fold higher occurrence in “At-Risk” cases, reflecting overt emotional distress and tearfulness. Sleep-related terms also demonstrate substantial discriminative power, with "asleep" (score: 0.206) showing 3.2-fold higher frequency in “At-Risk” interviews. These frequency differences indicate that participants spontaneously discuss sleep problems without explicit prompting, suggesting these symptoms are subjectively distressing and salient.

The term "basic" (score: 0.359, rank 2) shows 2.7-fold higher frequency in “At-Risk” cases, frequently appearing in self-critical statements about fundamental abilities such as "can't handle basic tasks" or "even basic stuff is overwhelming." This captures the negative self-evaluation and feelings of inadequacy characteristic of mental health challenges. Terms related to medical intervention also appear more frequently in “At-Risk” interviews, with the stemmed term "medic" (score: 0.217) showing 2.9-fold higher frequency. This encompasses references to medications, medical professionals, and healthcare contexts, indicating greater treatment engagement among participants classified as “At-Risk”.

A notable pattern emerges in the feature distribution: the top 7 most discriminative features (scores

0.206-0.500) are all “At-Risk”-associated, and “At-Risk” features generally exhibit larger frequency differentials than “Not At-Risk” features. For example, "couldn't" shows a 4.5-fold frequency difference, while the highest-ranked “Not At-Risk” feature "brother" shows only a 3.6-fold difference. This asymmetry suggests that potential mental health challenges manifest through more distinctive and consistent linguistic markers compared to psychological wellness. “Not At-Risk” participants demonstrate greater variability in language patterns—some discuss family extensively while others do not, some mention social activities while others focus on different topics. In contrast, “At-Risk” participants consistently employ language reflecting inability, emotional distress, sleep problems, and functional impairment, creating stronger and more reliable discriminative signals.

The convergence between discriminative features and established clinical symptom domains validates the TF-IDF approach's capacity to capture clinically meaningful indicators. The model bases classification decisions on linguistic patterns that mirror symptom domains assessed by standardized diagnostic instruments, rather than relying on spurious correlations or artifact patterns. Features reflecting functional impairment, emotional distress, sleep disturbance, and negative self-evaluation align with the symptom categories used in clinical assessment and the PHQ-8 questionnaire items that provided the ground truth labels for this study.

The feature importance analysis confirms that the SVM classifier bases decisions on clinically interpretable linguistic patterns aligned with mental health symptomatology. The dominance of functional impairment language, explicit emotional expression, sleep disturbance terms, and self-critical evaluation demonstrates that the TF-IDF representation successfully captures genuine indicators of potential mental health challenges. These findings establish confidence in the model's clinical validity and support its potential application in automated mental health assessment workflows.

4.3 End-to-End System Performance with ASR Transcripts

This section evaluates the complete pipeline by testing the optimal SVM classifier on ASR-generated transcripts of the same 28 test participants. This direct comparison isolates the impact of transcription quality on classification performance, addressing the central research question: whether the automated speech recognition maintains sufficient accuracy for clinically viable mental health assessment.

4.3.1 Performance Comparison between Manual and ASR transcripts

The SVM classifier was evaluated on ASR-generated transcripts of the identical 28 test set participants used for manual transcript evaluation in Section 4.2 . All other experimental conditions remained constant- the same trained model, identical preprocessing pipeline, and the same TF-IDF vectorization parameters. This ensures that any performance differences could be attributed only to transcription quality rather than methodological variations.

	Predicted: “Not At-Risk”	Predicted: “At-Risk”
Actual: “Not At-Risk”	15	7
Actual: “At-Risk”	1	5

Table 4.7.Confusion Matrix for SVM Classifier on ASR Transcript

Transcript Type	Accuracy (%)	F1-Score	"At-Risk" Precision	"At-Risk" Recall
Manual	75.0	0.76	0.45	0.83
ASR	71.4	0.74	0.42	0.83
Difference	-3.6	-0.02	-0.03	0.0

Table 4.8.Performance Comparison between Manual and ASR Transcripts

The ASR-generated transcripts achieved 71.4% accuracy compared to 75.0% for manual transcripts, representing a 3.6 percentage point absolute decrease. "At-Risk" recall remained stable at 83.3% for both transcript types (detecting 5 of 6 cases in both conditions). However, "Not At-Risk" recall decreased from 72.7% to 68.2% (correctly identifying 16 of 22 cases with manual transcripts compared to 15 of 22 with ASR transcripts). "At-Risk" precision remained relatively stable, declining only 0.02 points from 0.45 to 0.42. The weighted F1-score decreased modestly from 0.76 to 0.74, reflecting the overall balance of precision and recall across both classes.

The confusion matrices reveal distinct error patterns between manual and ASR conditions. In manual transcript classification, the model produced 1 false negative and 6 false positives. In ASR transcript classification, false negatives remained at 1 while false positives increased to 7. The analysis reveals that ASR transcription introduces a 3.6 percentage point reduction in classification accuracy, with the primary impact being reduced specificity (72.7% to 68.2% "Not At-Risk" recall) while sensitivity remained stable at 83.3%. Most classifications (71.4%, 20 of 28 participants) remained stable across transcript types, indicating substantial robustness of the TF-IDF and SVM pipeline to typical ASR errors.

4.3.2 Error analysis

7 cases out of 28 test participants (25%) were misclassified by at least one transcript type, these are participants 442, 441, 450, 469, 470, 445, 449, and 468. Participants 442, 441, 450, 469, 470, 445, and 449 were misclassified under both manual and ASR conditions, indicating these represent inherently challenging cases where linguistic patterns do not align clearly with PHQ-8 classifications.

Participant 468 represents the single case where ASR transcription directly caused classification failure. This participant, with a PHQ-8 score of 4 "Not At-Risk", was correctly classified as True Negative using manual transcripts but misclassified as False Positive using ASR transcripts with a confidence score of 0.63.

Analysis of the ASR transcript reveals specific transcription errors that created artificial linguistic patterns resembling "At-Risk" language. The ASR system introduced nonsense terms through recognition errors (e.g., "pdacius awkward" instead of "but now it's just awkward"), fragmented sentence structures that altered term co-occurrence patterns, and disrupted contextual qualifiers that normally mitigate negative affect terms. These errors disproportionately affected TF-IDF feature extraction by creating rare terms with maximum weights, isolating negative affect vocabulary from mitigating context, and increasing the density of first-person pronouns through sentence fragmentation.

The misclassification mechanism can be traced to ASR artifacts that mimicked "At-Risk" linguistic patterns learned during training. The fragmented sentence structure increased first-person pronoun density by approximately 40%, while negative affect term density increased by an estimated 67% due to loss of contextual qualifiers. The nonsense terms generated by recognition errors received maximum TF-IDF weights as rare vocabulary, inadvertently contributing to the "At-Risk" classification signal. These artificial patterns, combined with preserved but decontextualized stress-related vocabulary,

produced a feature profile that exceeded the classification threshold despite the participant's subclinical PHQ-8 score.

Despite the Participant 468 case, the system demonstrated substantial robustness for most classifications. Participants with strong “At-Risk” linguistic patterns (high frequency of discriminative features like "couldn't," "depress," "cry") maintained correct classification even when ASR introduced word-level errors, if core depression vocabulary was preserved. Similarly, participants with clear “Not At-Risk” profiles (absence of distress markers, presence of wellness indicators) typically remained correctly classified despite transcription imperfections. The vulnerability emerges primarily in borderline cases where classification decisions rest on subtle linguistic patterns or low-frequency features.

4.4 Discussion of Key Findings

This section concludes the experimental findings to address the main research topic, usage of Automatic Speech recognition for practical mental health screening from audio. The study tested a complete pipeline combining the NVIDIA NeMo ASR systems with an SVM classifier using TF-IDF features, and the results show both where the system works well and where it struggles.

4.4.1 ASR Performance and Clinical Speech Characteristics

The NeMo ASR system performed well on clinical interview, achieving 86.3% mean accuracy across 28 participants. Most sessions (85.7%) exceeded 80% accuracy. Performance patterns were predictable, as clear, steady speech produced accuracies above 90%, while rapid, emotional, or hesitant speech reduced accuracy to as low as 70.2%. Clinical terms like “PTSD” were often broken into individual letters, showing that the system struggles with medical abbreviations.

The error analysis revealed an important pattern, where the character error rate (7.4%) stayed consistently lower than word error rate (13.2%). This means transcription errors often produced partial words or spelling variations rather than compeletely different words. This partial preservation explains why the classification system remained robust- clinically meaningful content was usually kept despite surface-level mistakes.

4.4.2 Linguistic Classification and Clinical Validity

On manual transcripts, the SVM classifier achieved 75% accuracy, matching the PHQ-8 screening instrument’s own performance. The model showed balanced results: 83.3% recall for “At-Risk” cases and 72.7% recall for “Not At-Risk” cases. The 2:1 class weighting successfully prioritized finding “At-Risk” cases without creating too many false positives, which is appropriate for a screening tool.

The feature analysis confirmed that classification decisions were based on clinically meaningful patterns. the top features: “couldn’t” (functional problems), “depress” (direct symptom mention), “cry” (emotional expression). and “asleep” (sleep problems). All the main features match symptom areas that mental health screening tolls assess. This shows the model detects real symptom language rather than random patterns. “At-Risk” features also showed bigger frequency differences than “Not At-Risk” features, suggesting that language linked to mental health challenges is more distinctive and consistent.

4.4.3 End-to-End System Performance

When the optimal model was tested on ASR transcripts, accuracy dropped by 3.6% to 71.4%, but “At-

Risk" recall stayed at 83.3%. This means the system still detects potential cases despite transcription errors. The main change was in specificity "Not At-Risk" recall feel from 72.7% to 68.2%, showing a shift toward higher sensitivity.

Importantly, 71% of classifications (20 of 28 participants) remained correct under both conditions, demonstrating that the TF-IDF and SVM combination handles typical recognition errors well. Among the 8 cases where at least one condition produced misclassification, 7 were misclassified under both manual and ASR conditions, indicating inherent difficulty rather than ASR-induced failure. Only one case (Participant 468) was misclassified purely because of ASR errors, where broken sentences and isolated negative words created artificial "At-Risk" patterns. This shows that the system works best for participants with clear language patterns but struggles with borderline cases where subtle context matters.

4.4.4 Clinical and Practical Implications

These results show that fully automated, speech-based mental health screening is feasible for initial assessment. "At-Risk" recall stayed stable across both transcript types, meaning the system can reliably identify people who need further evaluation even with imperfect transcription. The high "Not-At-Risk" precision (94.1%) means negative predictions are very trustworthy, which is useful for efficient clinical workflows.

However, the error patterns show specific considerations. ASR errors don't happen randomly—they create systematic distortions that can produce false positives. Participant 468 demonstrates how transcription errors can create artificial feature patterns that look like "At-Risk" language even when PHQ-8 scores are low. This problem appears strongest in borderline cases where classification depends on subtle language patterns.

The performance seen in this study 71.4% accuracy with 83.3% "At-Risk" sensitivity and one ASR-caused misclassification among 28 cases suggests automated pipelines could work as preliminary screening tools where sensitivity is most important and false positives are acceptable for the sake of scalability. The finding that 20 of 28 classifications remained correct across both transcript types shows the approach works particularly well for cases with distinctive language patterns, though borderline cases might need additional checks.

4.4.5 Study Limitations

This study used a small and real clinical data set, which limits how broadly these findings apply. The data set was imbalanced, which caused biased behavior of the model, even different weighting method was applied. Moreover, the study used a general-purpose ASR model without training on clinical speech, which may underestimate what systems trained on therapy conversations could achieve. The TF-IDF features capture word frequency but miss semantic context and negation, potentially overlooking nuanced language patterns. The binary classification based on PHQ-8 threshold (score $\geq$10) does not capture severity levels or distinguish between different mental health conditions. Ground truth labels came from PHQ-8 self-reports, which have their own diagnostic limits (74% sensitivity, 71% specificity compared to clinical interviews). This creates a performance ceiling that limits maximum achievable accuracy.

Chapter 5

5 Conclusion and Future Work

5.1 Conclusion

This research evaluated whether automated speech recognition can be used in mental health screening from audio. The study combined NVIDIA NeMo ASR system with TF-IDF feature extraction and SVM classification to create a complete end-to-end pipeline out of different models which were tested.

The NeMo system achieved 86.3% transcription accuracy on clinical interview audio, with character-level errors (7.4%) remaining consistently lower than word-level errors (13.2%). This pattern indicates that transcription errors typically produced partial words or spelling variations rather than completely different words, preserving semantic content crucial for classification. On manual transcripts, the SVM classifier achieved 75% accuracy with 83.3% “At-Risk” recall, demonstrating balanced performance across both classes. When tested on the complete audio-to-assessment pipeline, accuracy decreased by 3.6 percentage points to 71.4%, while “At-Risk” recall remained stable at 83.3%. Critically, only one case among 28 was misclassified solely due to audio processing errors, with 71.4% of classifications remaining correct across both conditions.

These findings demonstrate that mental health screening directly from audio is feasible with current ASR technology. The stable sensitivity (83.3%) combined with high “Not At-Risk” precision (94.1%) creates a suitable profile for preliminary screening applications. The modest 3.6 percentage point accuracy reduction represents an acceptable trade-off for eliminating manual transcription requirements and enabling scalable deployment. Mental health screening can now be automated directly from audio recordings, enabling implementation in telehealth consultations, mobile health check-ins, and automated monitoring systems where audio is the natural input modality.

5.2 Future Work

The most important next step is obtaining high-quality clinical audio datasets. the current study utilized one of the few publicly available mental health audio datasets on the internet, accessed with official permission from the data providers. [37] [38]. However, finding clinical audio datasets that combine therapeutic conversations with manual transcripts remains extremely challenging, as most such data is restricted due to privacy concerns or simply does not exist in sharable formats. Future work would benefit significantly from larger collections of actual therapeutic conversations with both clean audio recordings and manually verified transcripts, spanning diverse accents, demographics, and recording conditions. Better clinical data would enable several key advances. First, existing ASR models like Nvidia NeMo could be fine-tuned specifically on therapeutic speech patterns, improving their handling of medical terminology and emotionally modulated speech. Second, richer datasets would allow moving from TF-IDF to contextual language models like BERT that better capture semantic nuance and negation, leading to more accurate text-based analysis.

Also, it opens a dual pipeline architecture. One pipeline would analyze the transcript using advanced language models to extract linguistic patterns. The other would analyze the raw audio signal directly, extracting acoustic-prosodic features through deep learning models working on audio spectrograms. as audio contains, valuable information that text cannot capture - pitch patterns, speech rate, pauses,

energy levels, vocal tension. The two pipelines would then combine their assessments for a final prediction. When transcription errors occur, acoustic features could compensate. When someone's words seem fine, but their vocal patterns suggest distress, the audio analysis could detect it. This kind of multimodal system would be substantially more robust than the current approach. As clinical audio datasets become available and these dual-pipeline methods mature, audio-based mental health assessment could expand access to screening, reduce clinician workload in resource-limited settings, and enable continuous monitoring through audio recordings in real-world healthcare environments.

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8 Annex

8.1 Declaration on the Use of Generative Artificial Intelligence

☐ **I have not used** any generative AI tools.

☒ **I have used generative AI tools.** I have verified the content generated by AI, ensured the accuracy of the outputs, and properly indicated each instance of use in the table below.

Usage type	**Name of Generative AI Tool(s)**	**Affected Sections** (chapter, page number, reference)	**Estimated Proportion of Use** (per usage type)
Literature Review	DeepSeek, Claude	Chapter 1- Introduction, Chapter 2- Related Work	15%
Brief Summary of the Prompt	"I have identified key findings and the research gap from selected papers. Here is the logical flow and content for the Related Work chapter. Please rewrite this to ensure a formal academic tone, correct grammar, and clear structure"		
Program Code Generation	DeepSeek, Claude	Chapter 3- Methodology	30%
Brief Summary of the Prompt	"I need to preprocess the DAIC-WOZ data: 1) Align manual transcripts with audio from t=120s, filtering for only participant speech. 2) Implement a participant-level data split to prevent leakage. 3) Implement Grid Search with stratified cross-validation for model tuning. I have the logic/design; please help with the Python/pandas/scikit-learn implementation".		
Generating New Ideas or Solution Proposals			
Brief Summary of the Prompt			
Creating an Outline (text structure, bullet points)	DeepSeek, Claude	Chapter 3- Methodology Chapter 4- Experimentla Results and Perfromance Evaluation	5%
Brief Summary of the Prompt	"I have my draft notes and the logical flow for the Methodology/Results chapters. Please help organize this into a formal, hierarchical structure with clear section and subsection titles to improve the organization and academic presentation".		
Creating Text Blocks	DeepSeek, Claude	Chapters 1-4	25%
Brief Summary of the Prompt	"I have written this paragraph, but the wording is too informal/repetitive. Please rewrite it in formal academic style while preserving the original meaning".		
Generating Images for Illustrative Purposes			

Brief Summary of the Prompt			
Data Visualization, Generating Charts Based on Data Points			
Brief Summary of the Prompt			
Preparing a Presentation			
Brief Summary of the Prompt			
Other (please specify)			
Brief Summary of the Prompt			
Aggregated Percentage Value (for the core part of the task)			**18.75%**

Brief Textual Justification of the Aggregated Value:
Generative AI was used for language polishing, structural refinement of specific technical chapters, and technical implementation support**.** All theoretical content, research design, methodological definitions, and logical structure originated from me. I used AI to:

1. improve the academic tone and clarity of my written drafts (Text Blocks),
2. create a formal hierarchical structure (section/subsection titles) for the Methodology and Results chapters based on my notes
3. overcome specific implementation hurdles in Python for data preprocessing and model tuning after designing the approach. Every output was reviewed, validated, and integrated.

The tools served strictly as assistants for execution and expression—not as sources of intellectual content—and final responsibility for the thesis lies entirely with me.